

Advanced Data Processing Report

File: House_PriceReg.csv

Original Data Shape: (506, 19)

Numeric Columns: ['price', 'crime_rate', 'resid_area', 'air_qual', 'room_num', 'age', 'dist1', 'dist2', 'dist3', 'dist4', 'teachers', 'poor_prop', 'n_hos_beds', 'n_hot_rooms', 'rainfall', 'parks']

Categorical Columns: ['airport', 'waterbody', 'bus_ter']

Selected Processing Steps: ['Outlier_Handling', 'Missing_Values', 'Categorical_Encoding', 'Dummy_Variables', 'Duplicate_Removal', 'Feature_Scaling', 'Inconsistent_Data']

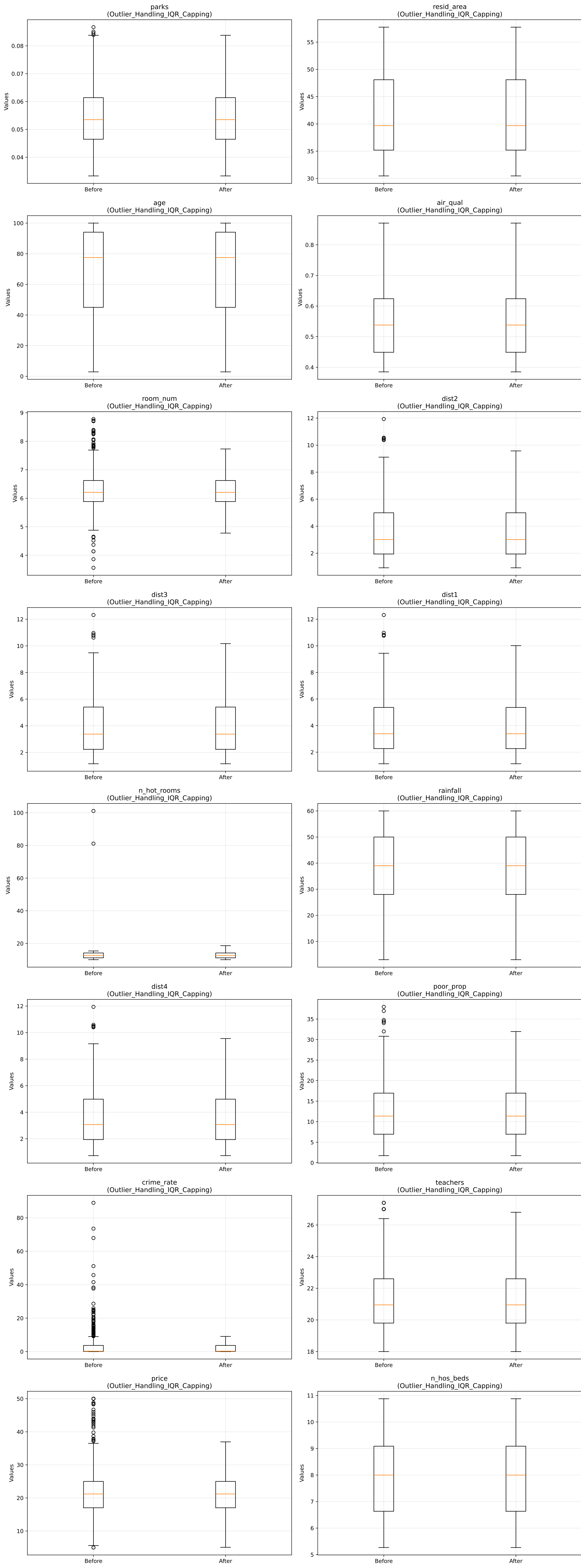
Processing Date: 2025-10-10 01:30:57

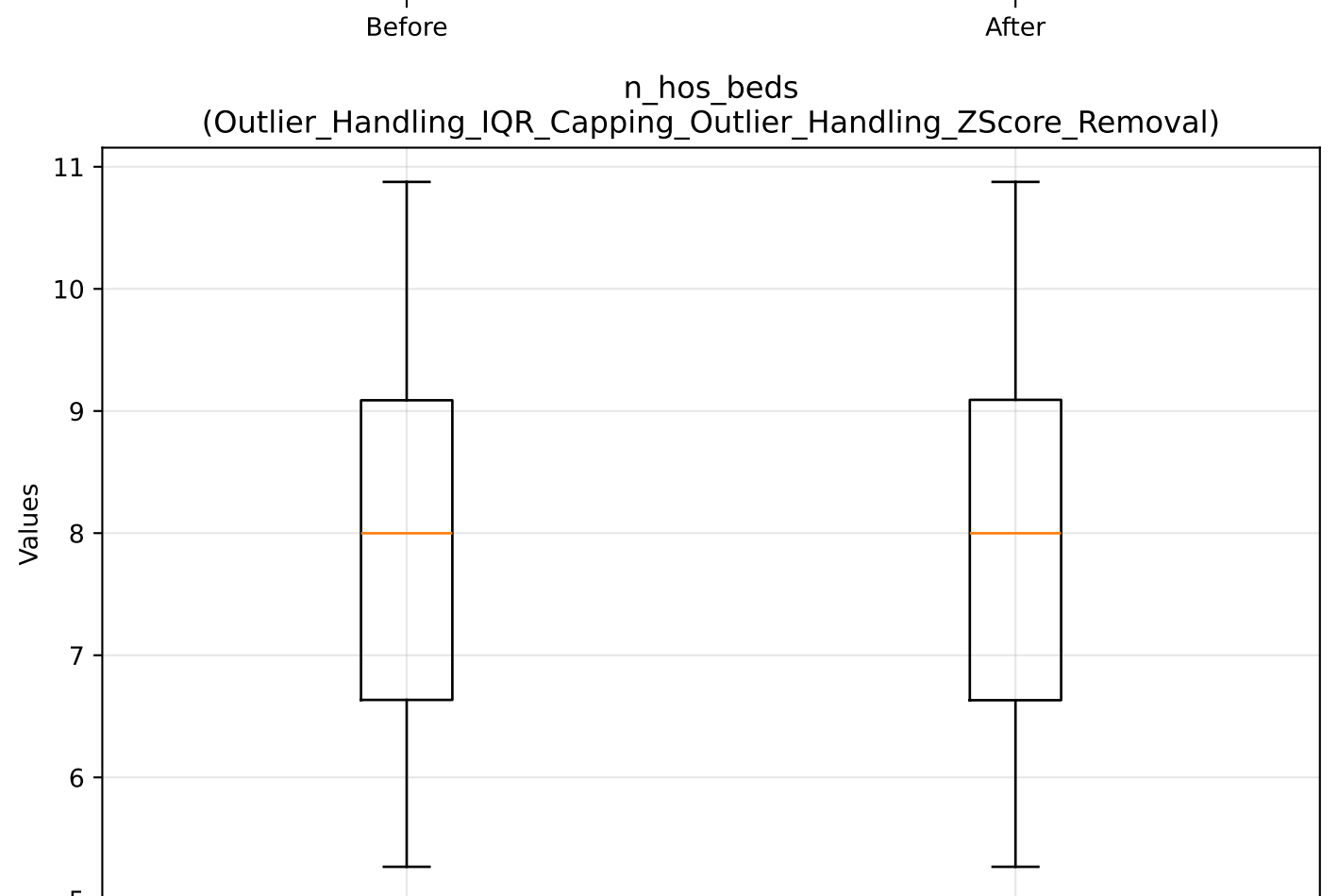
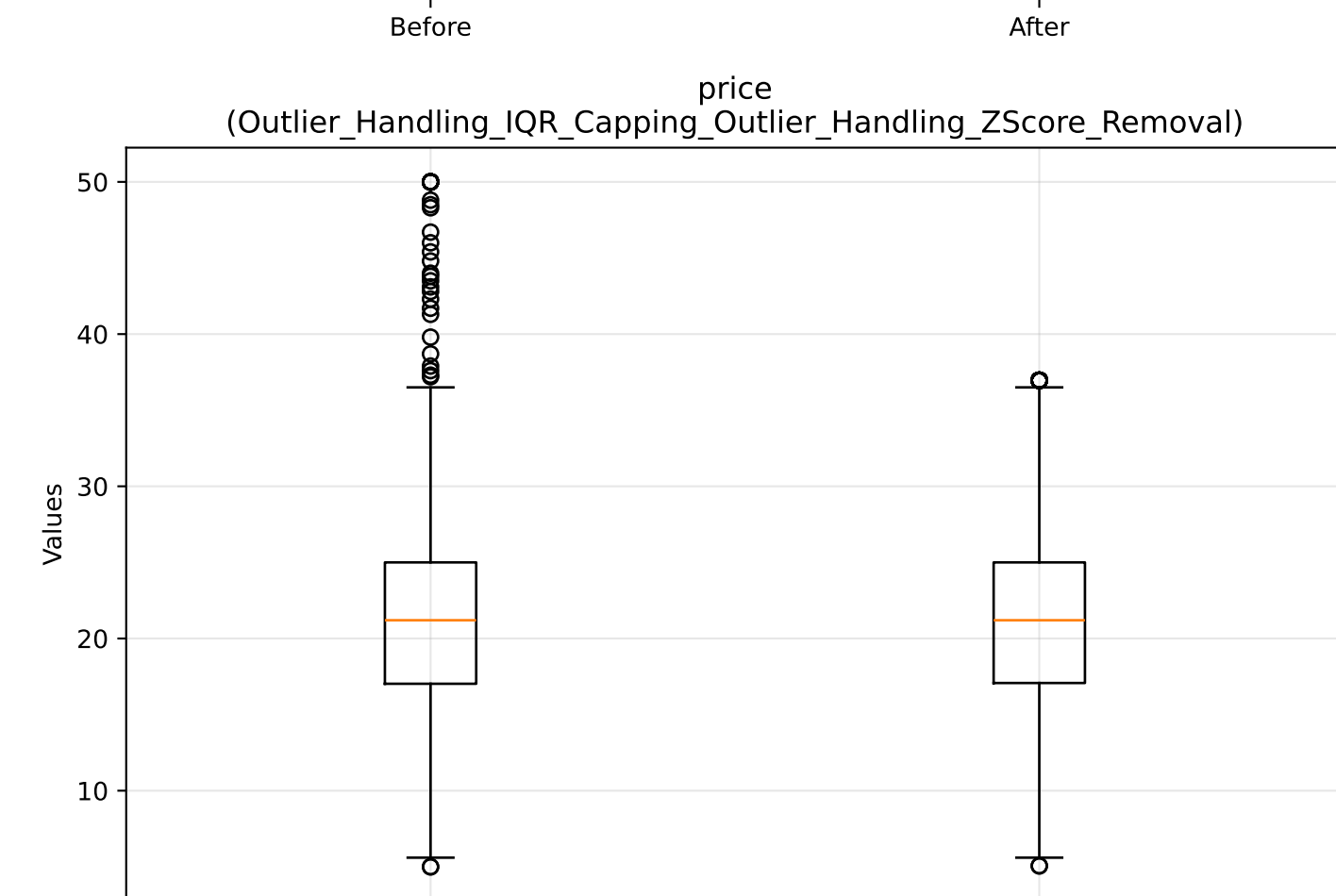
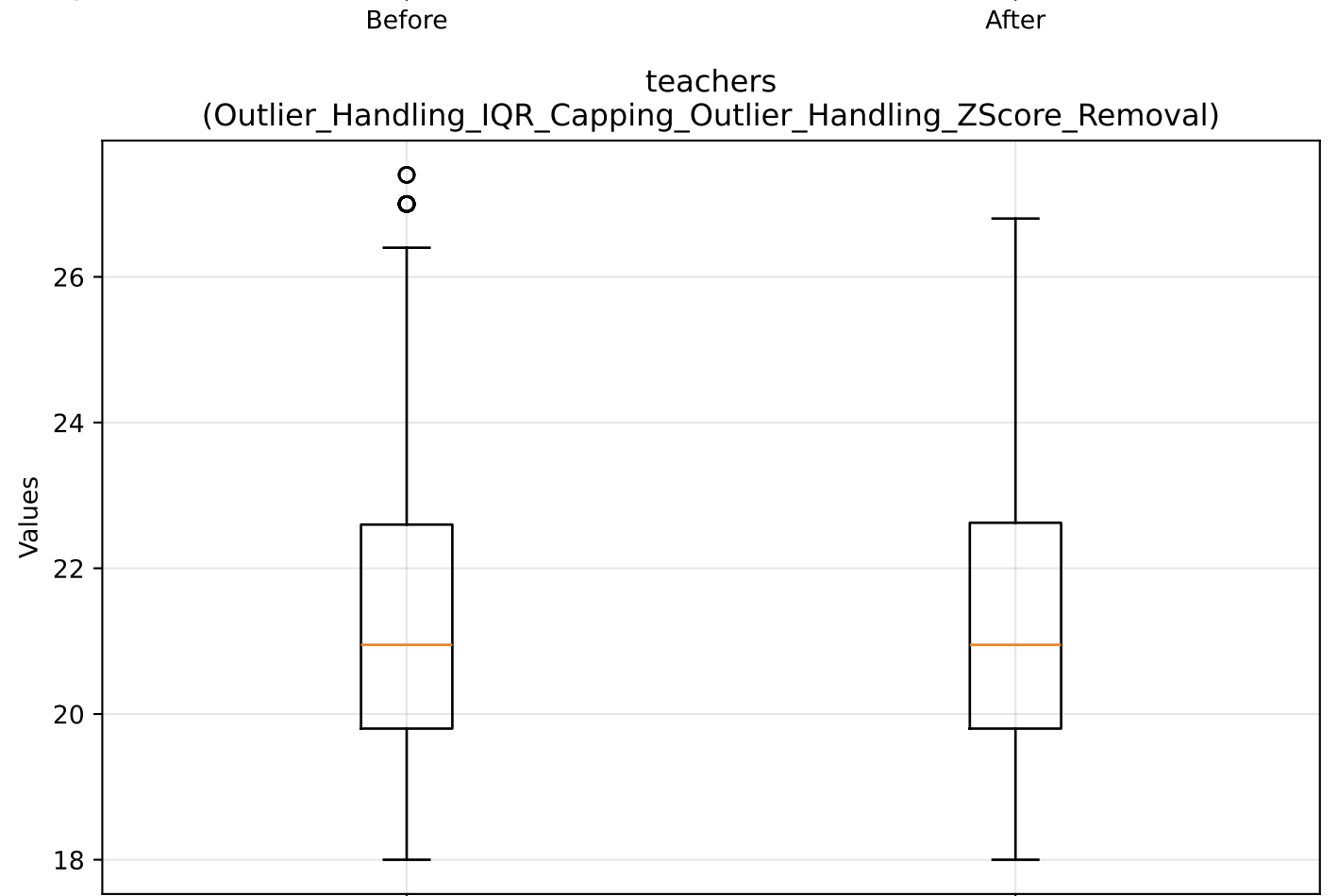
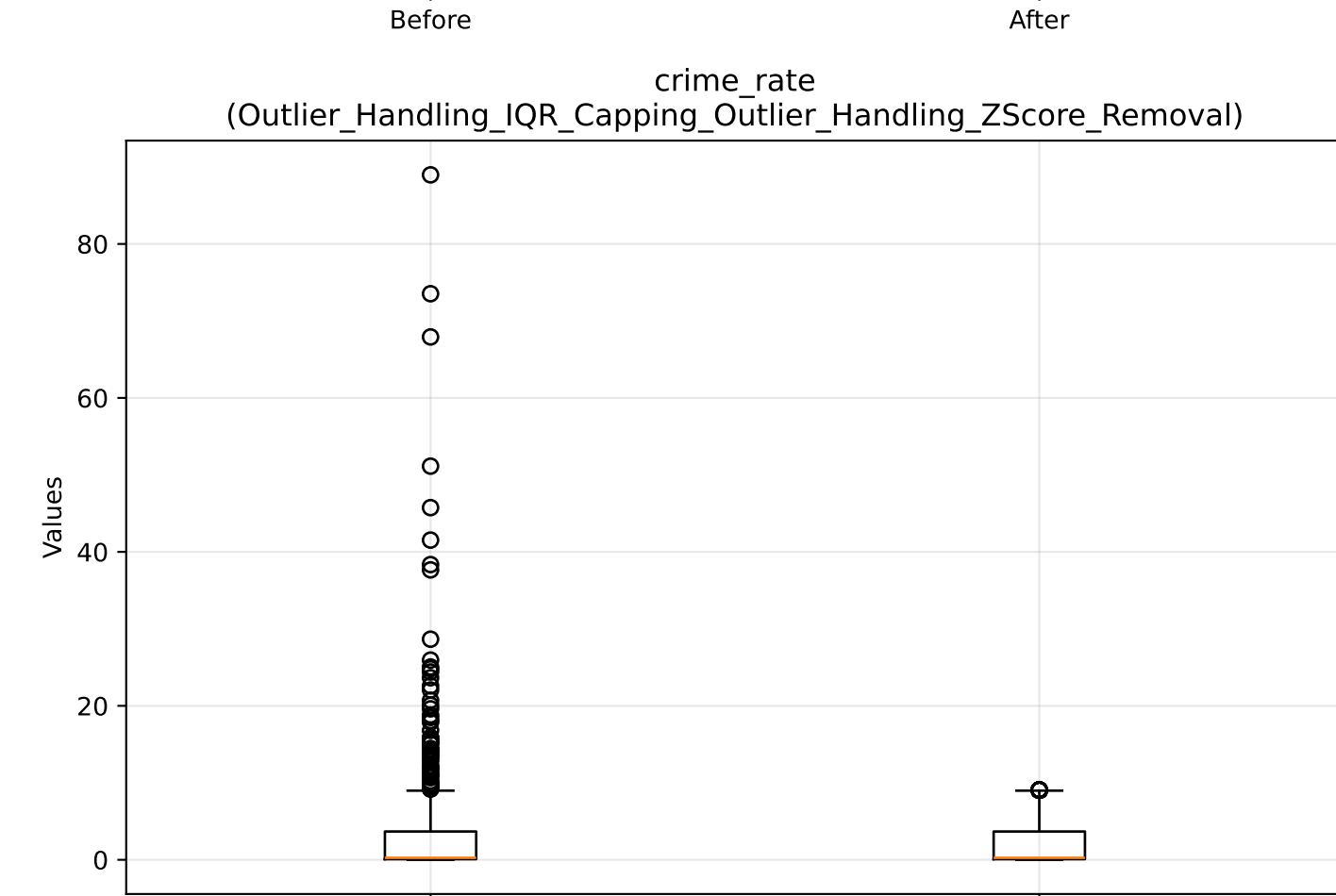
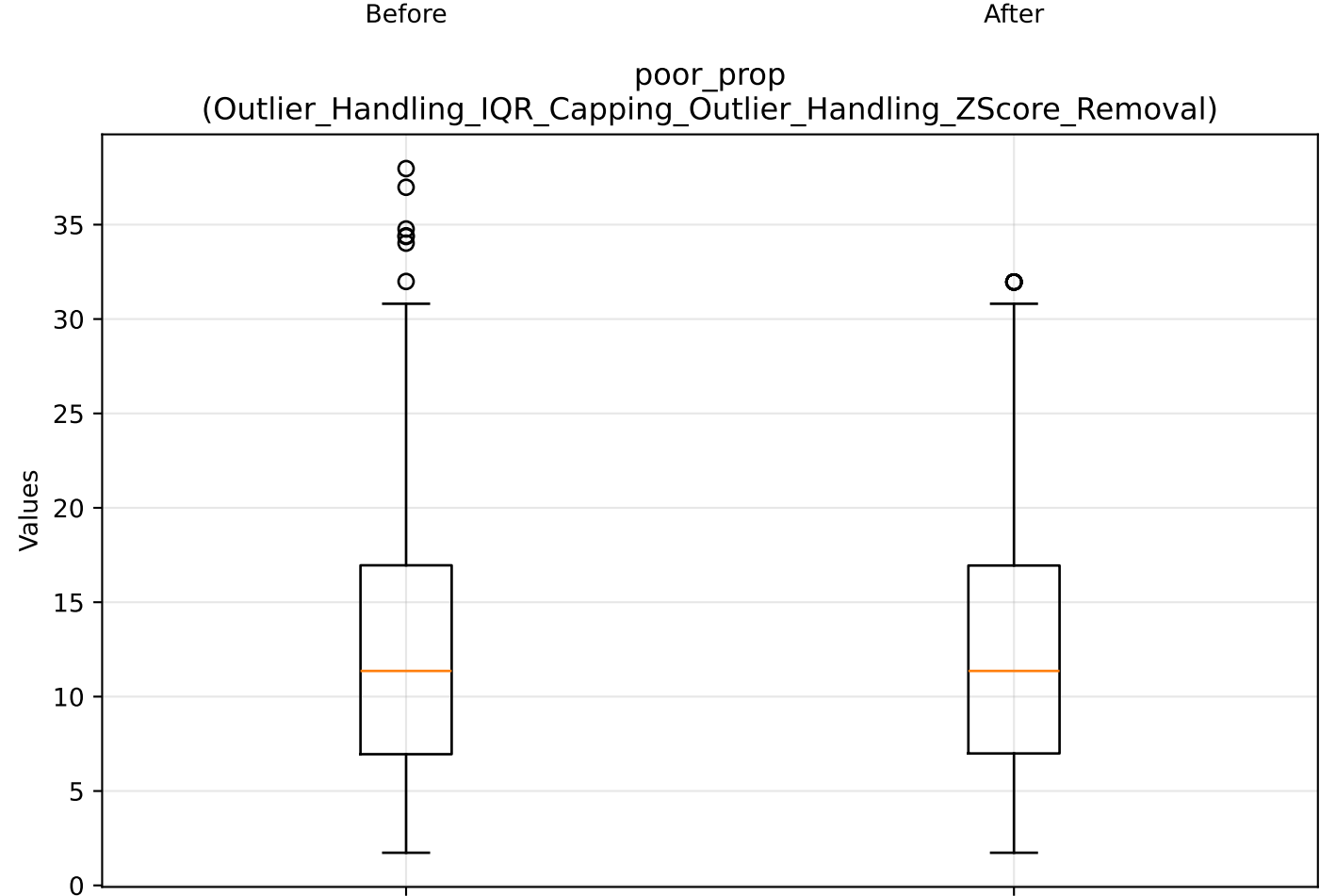
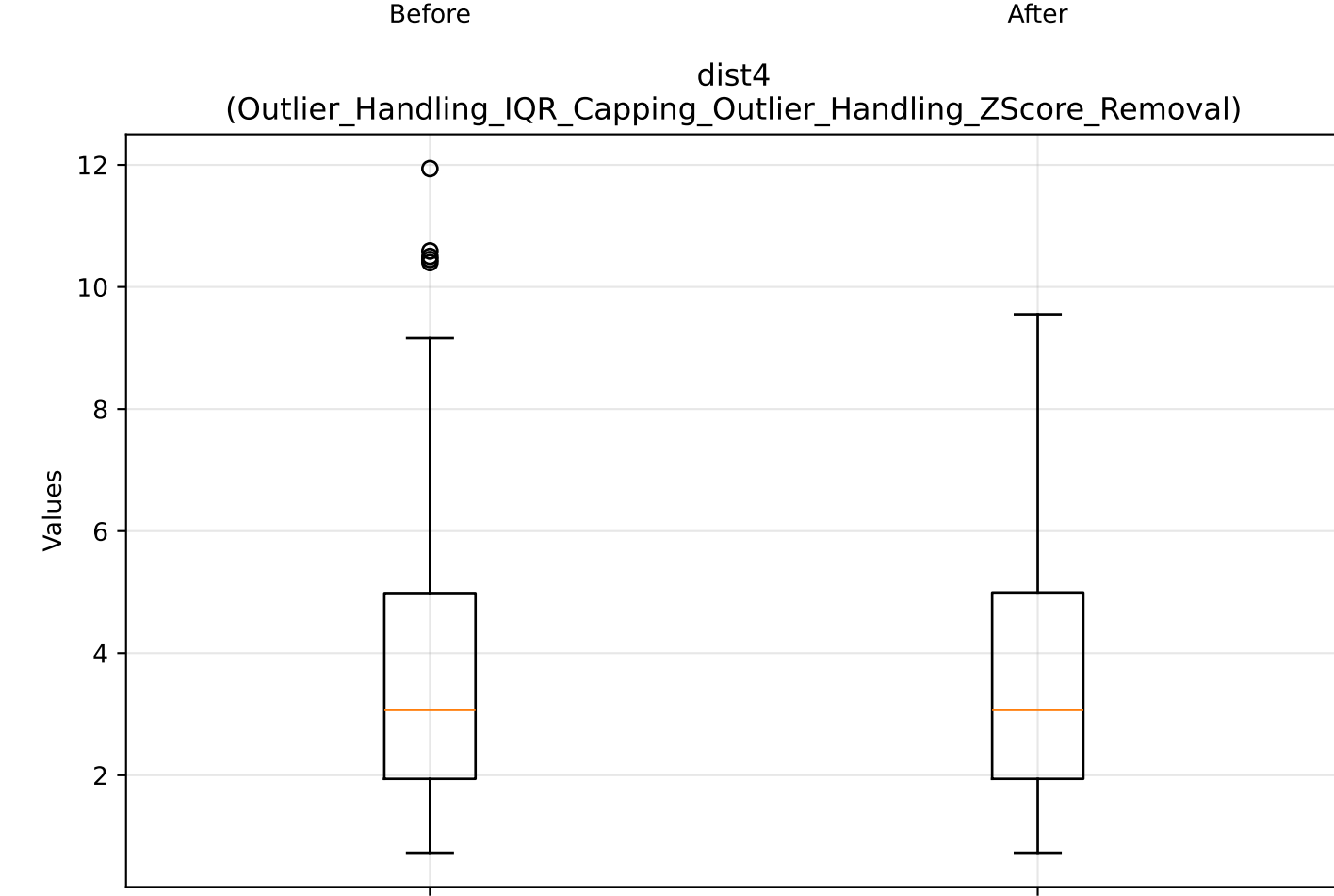
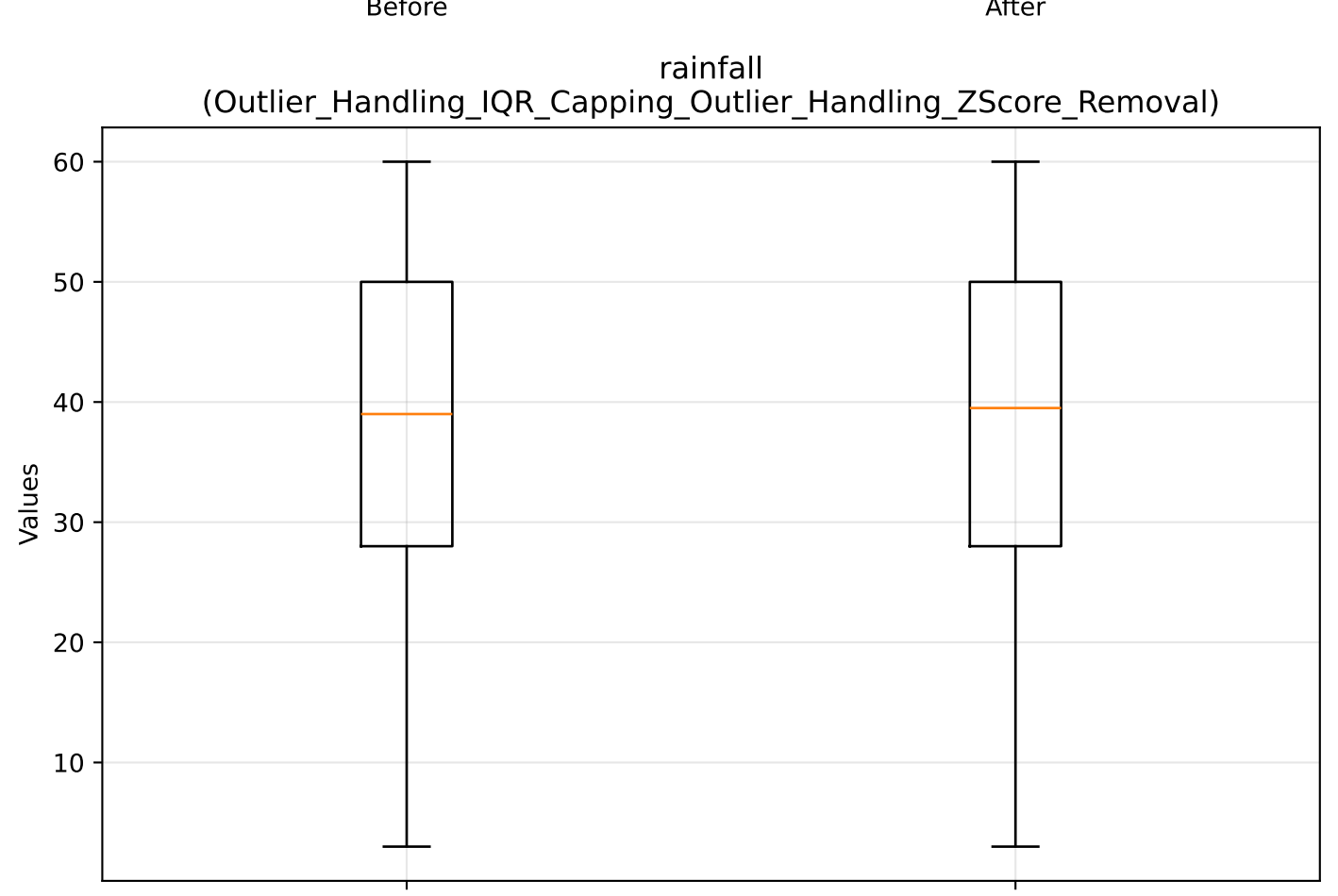
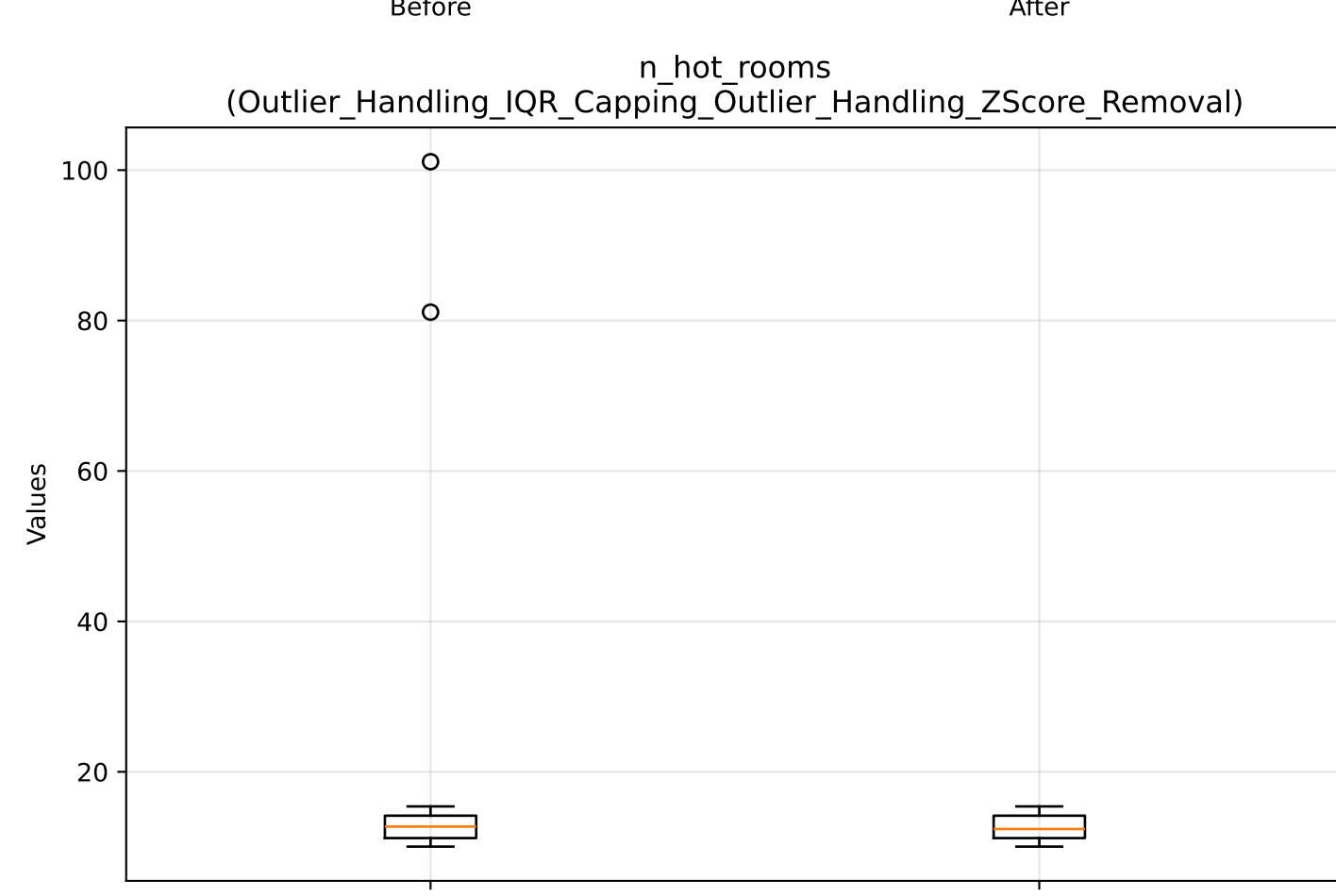
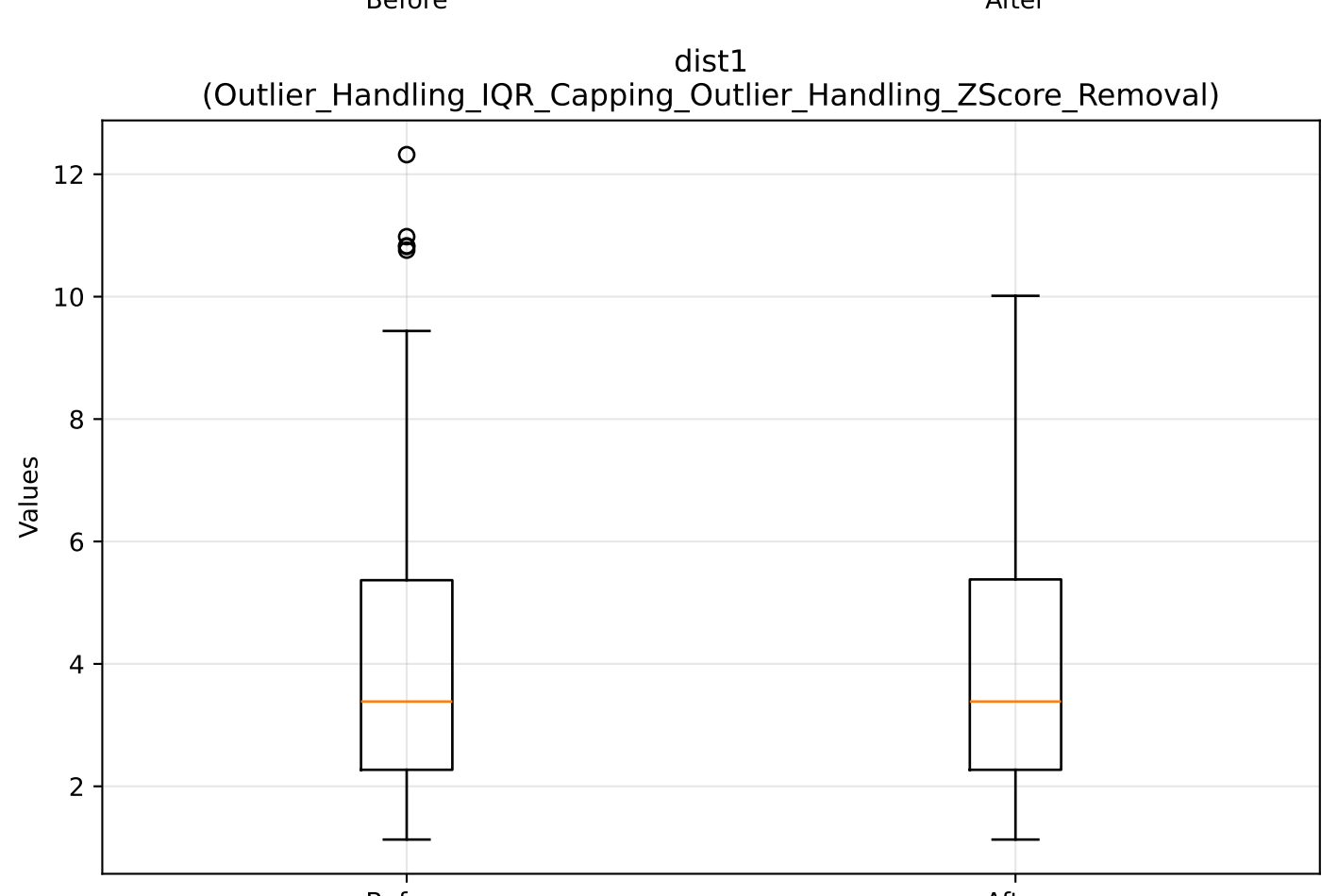
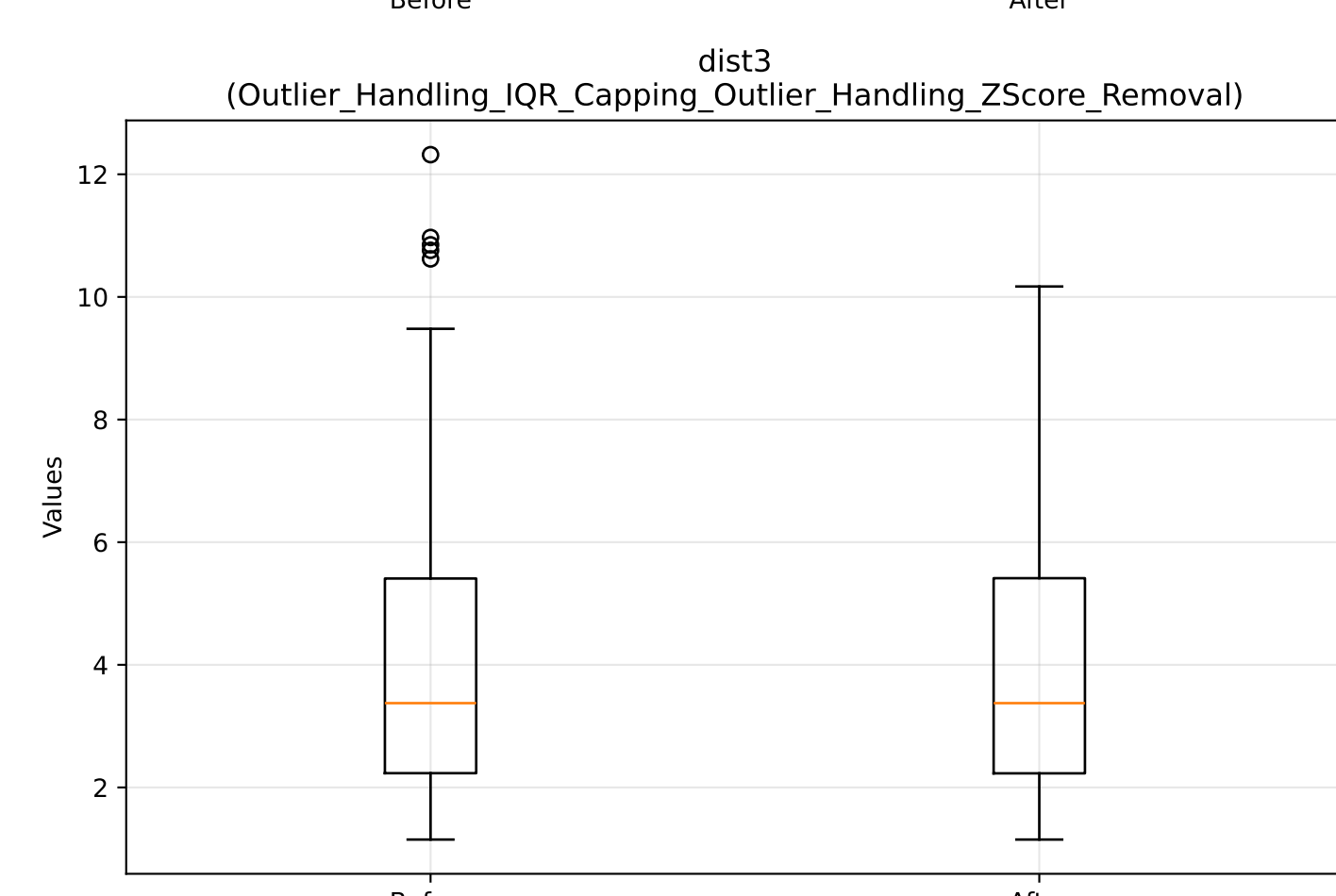
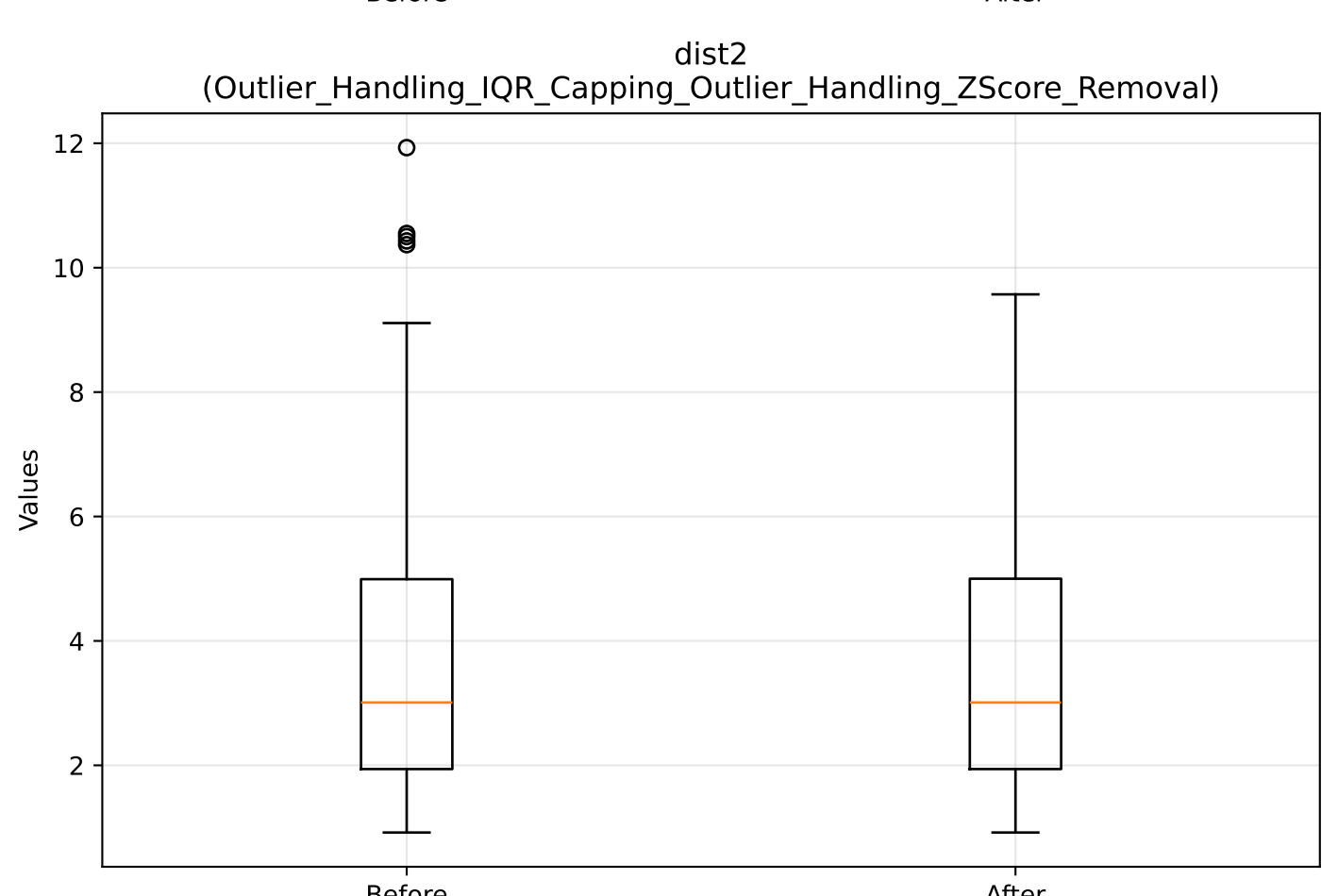
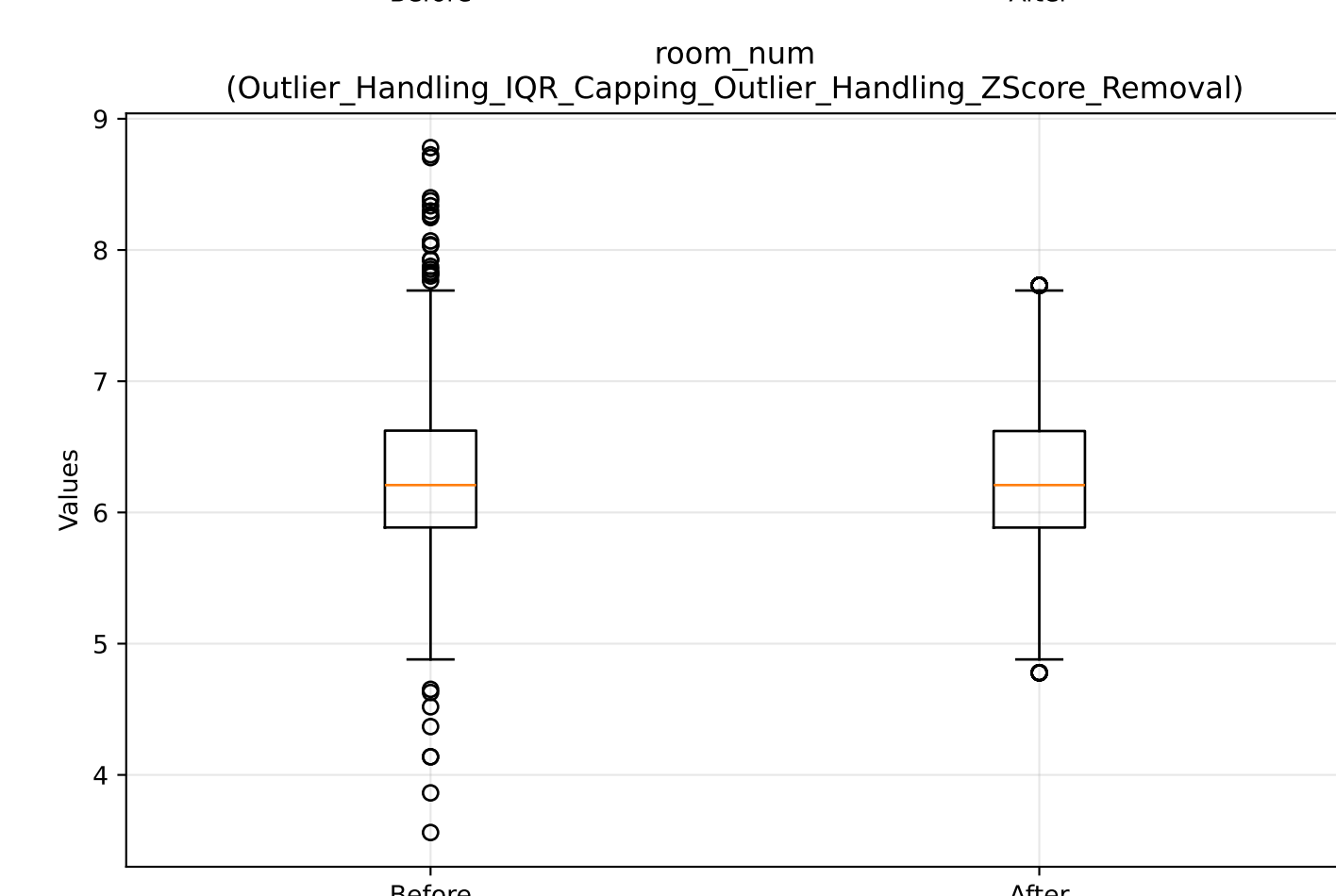
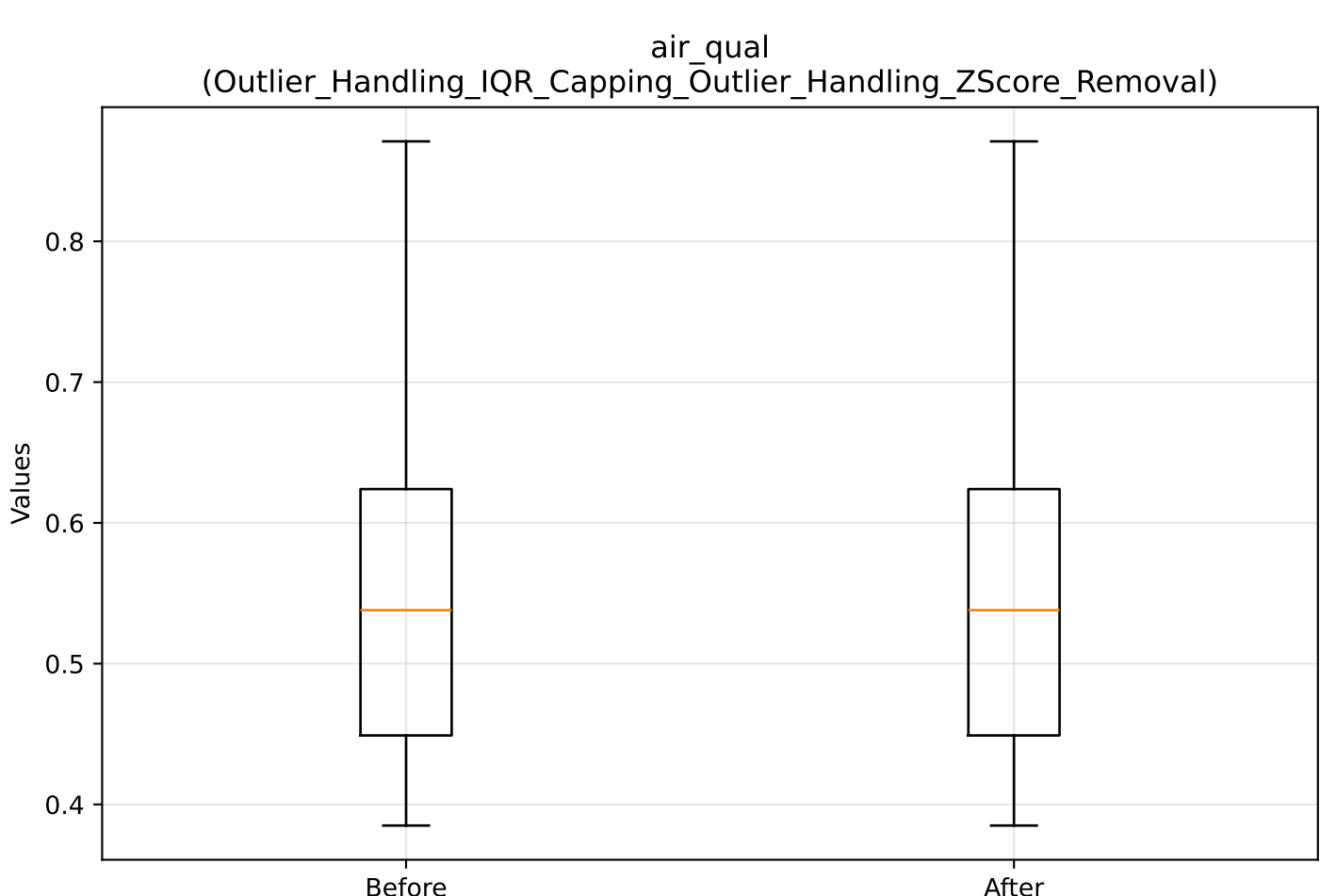
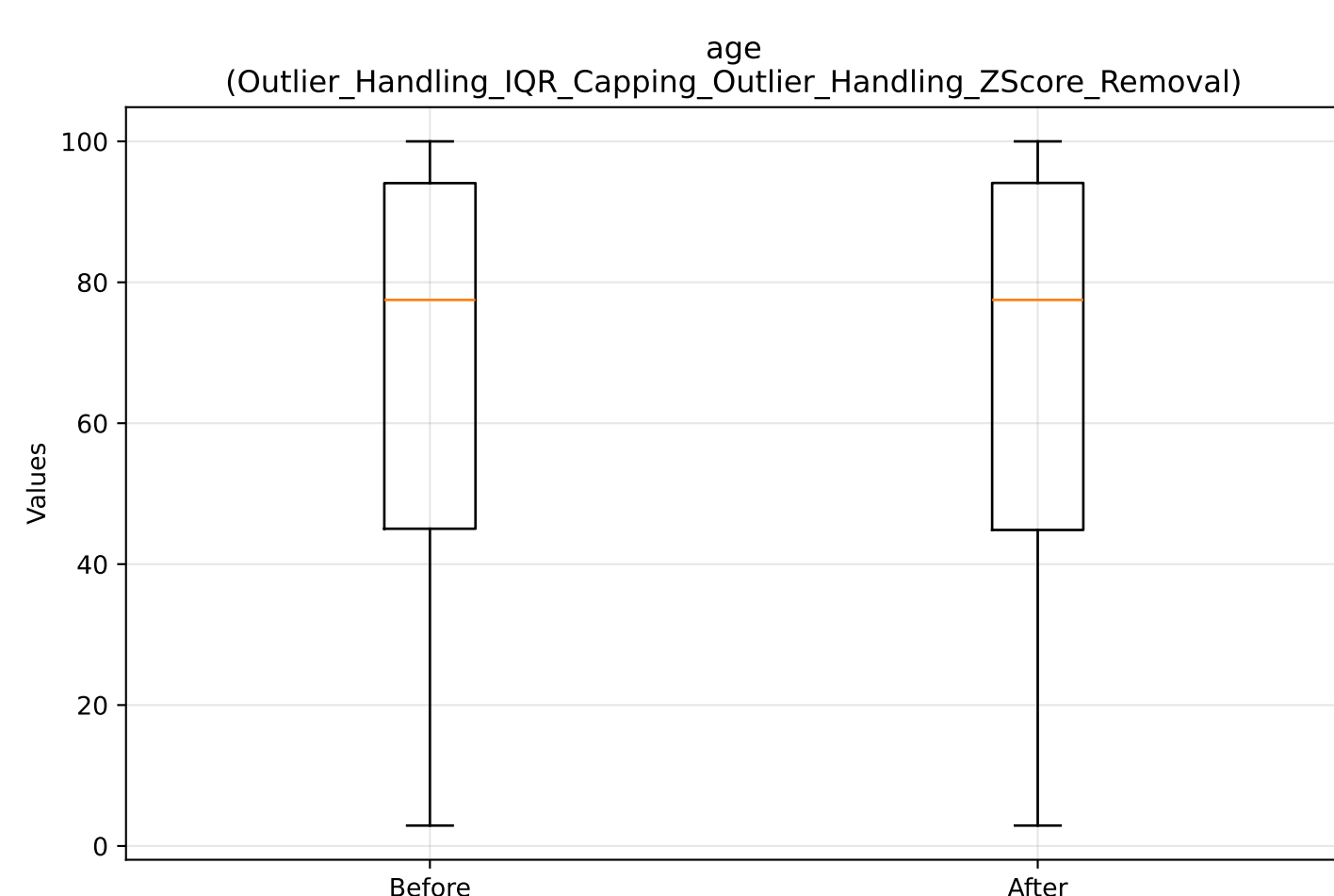
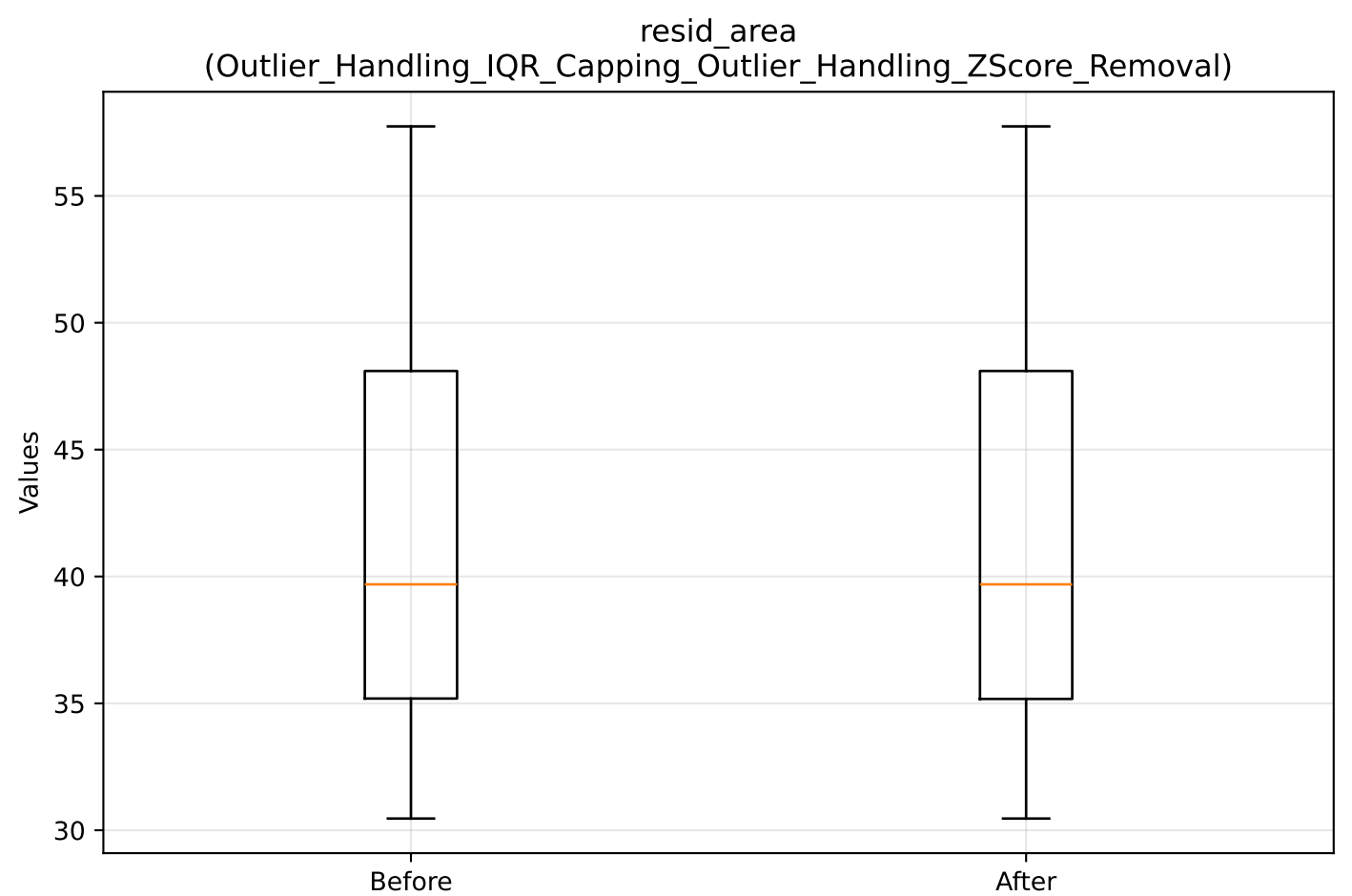
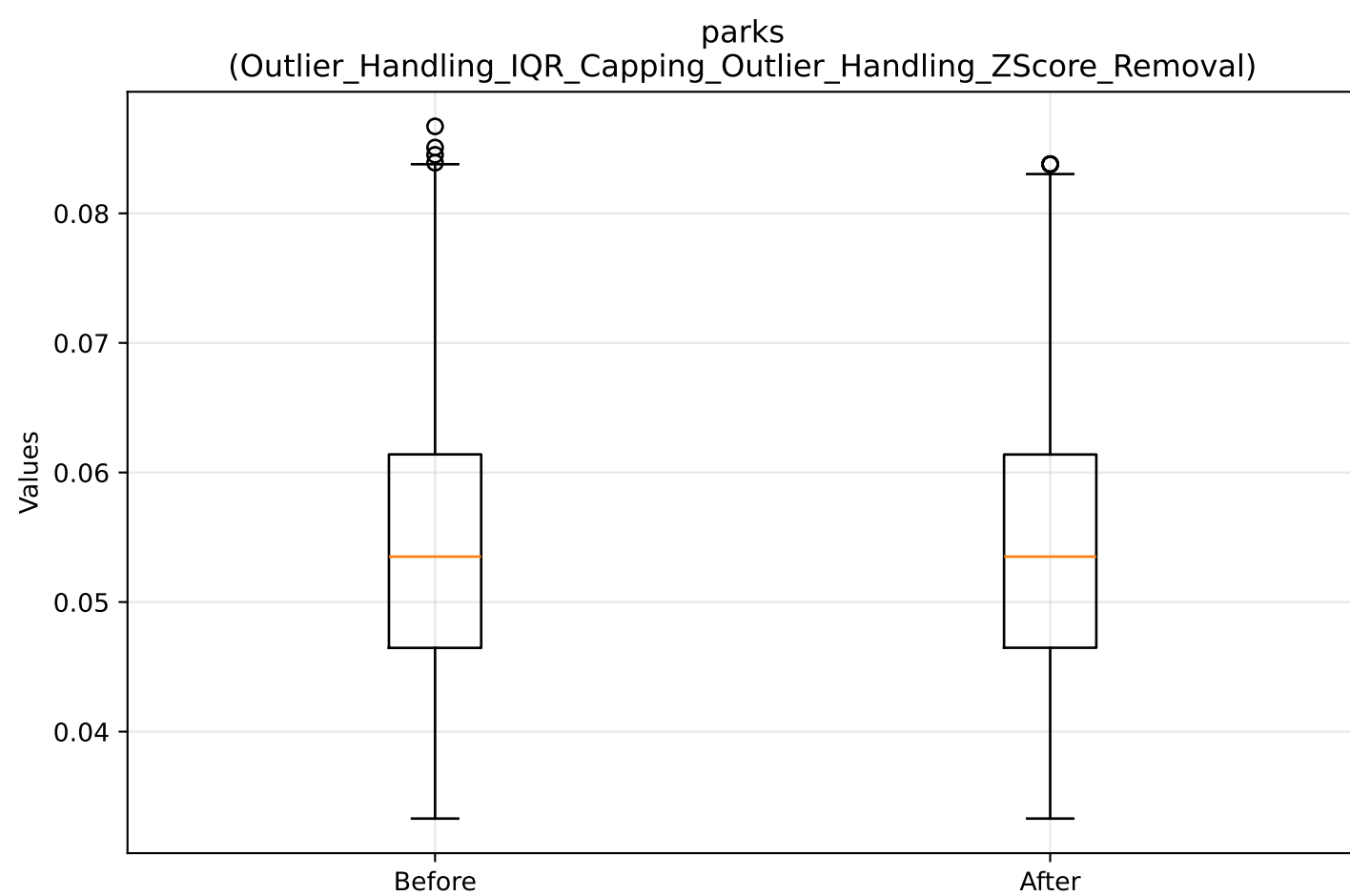
Processing Steps Applied:

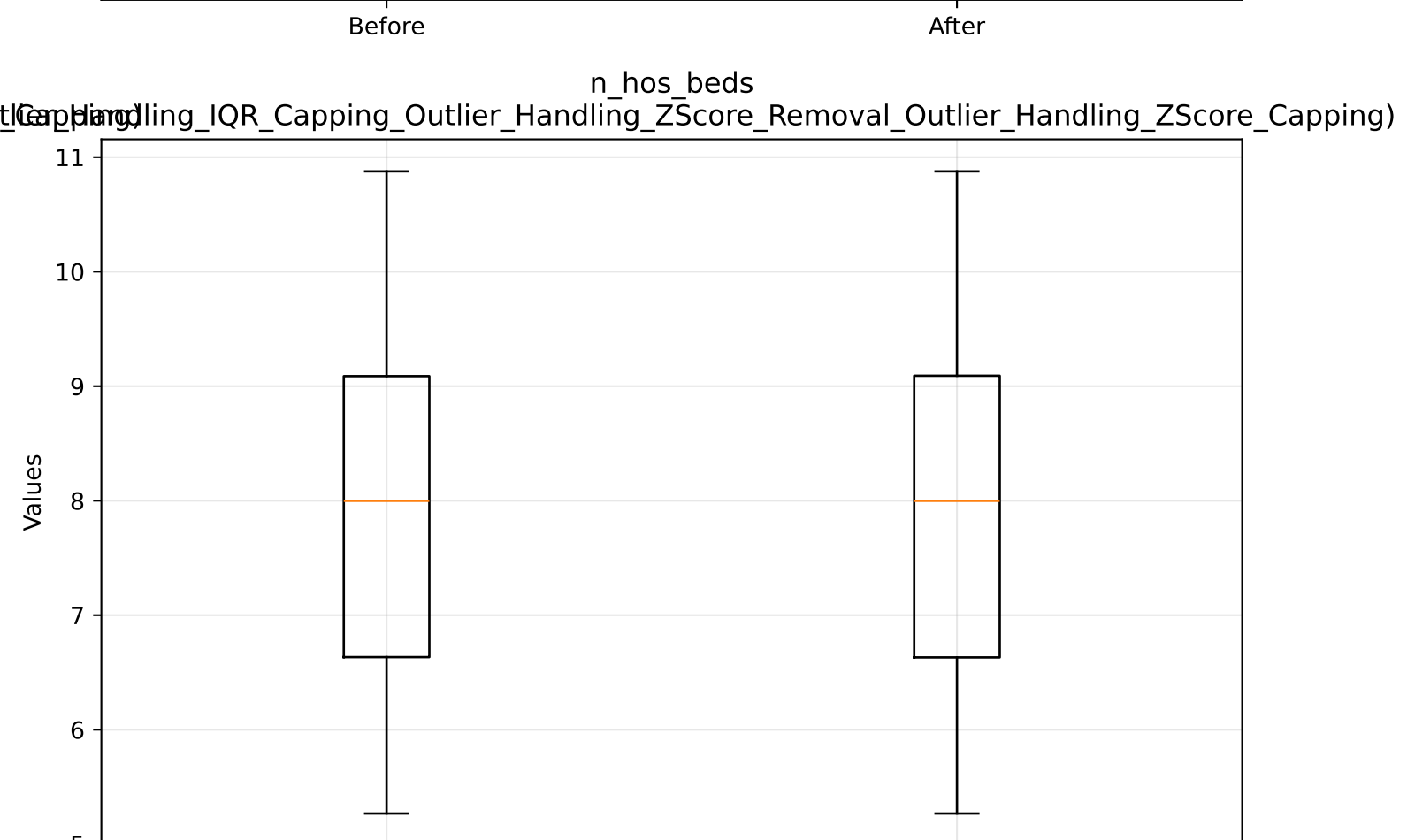
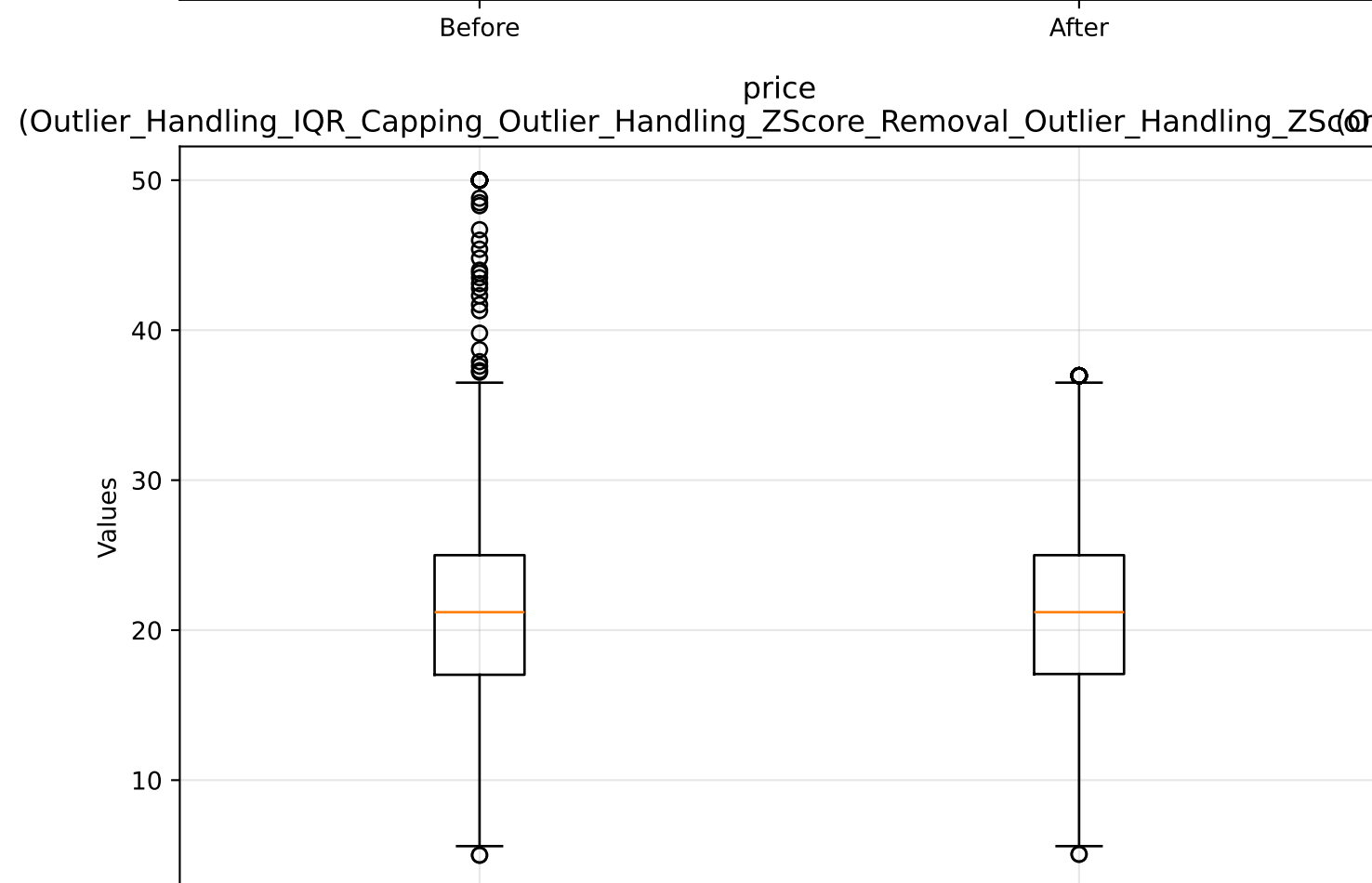
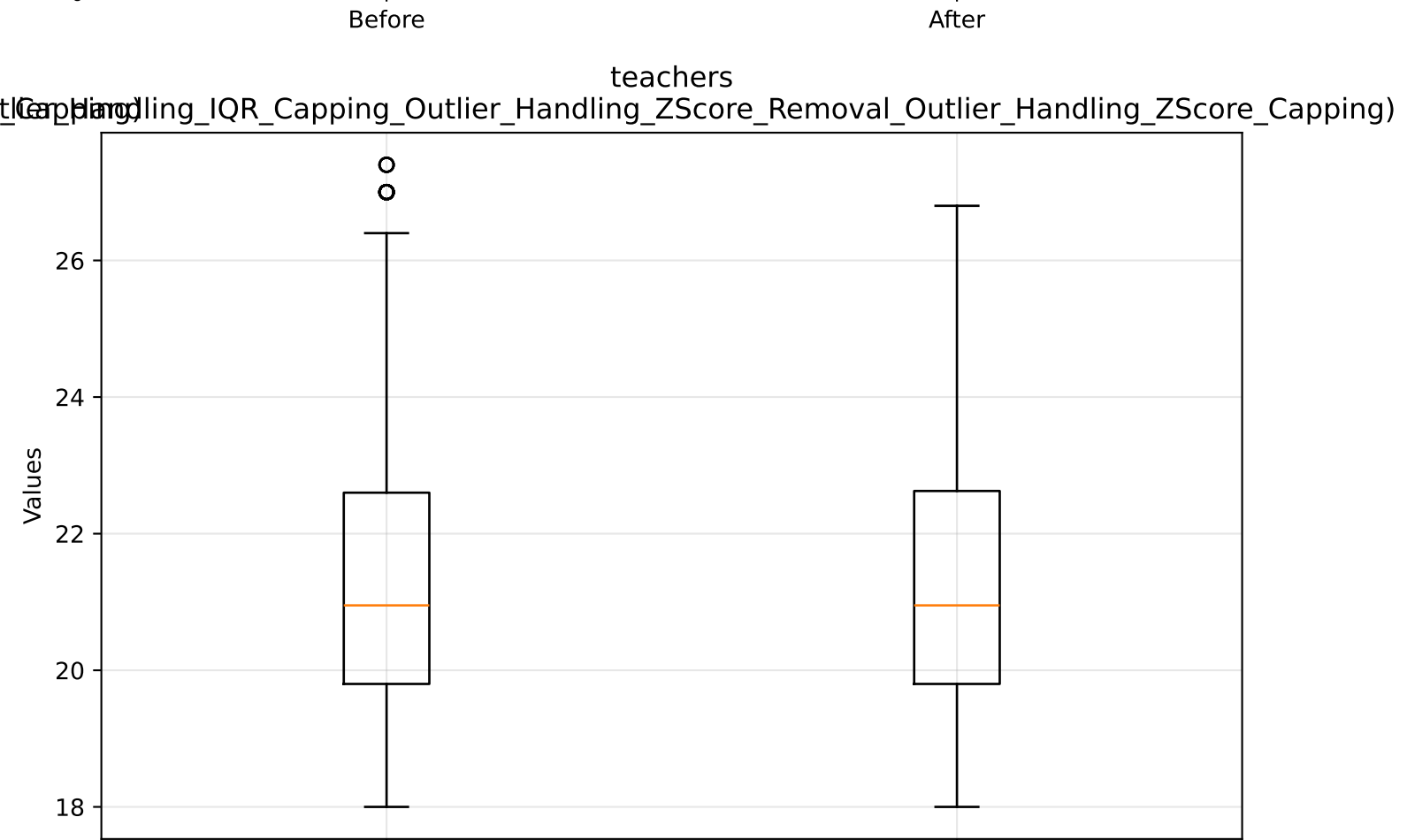
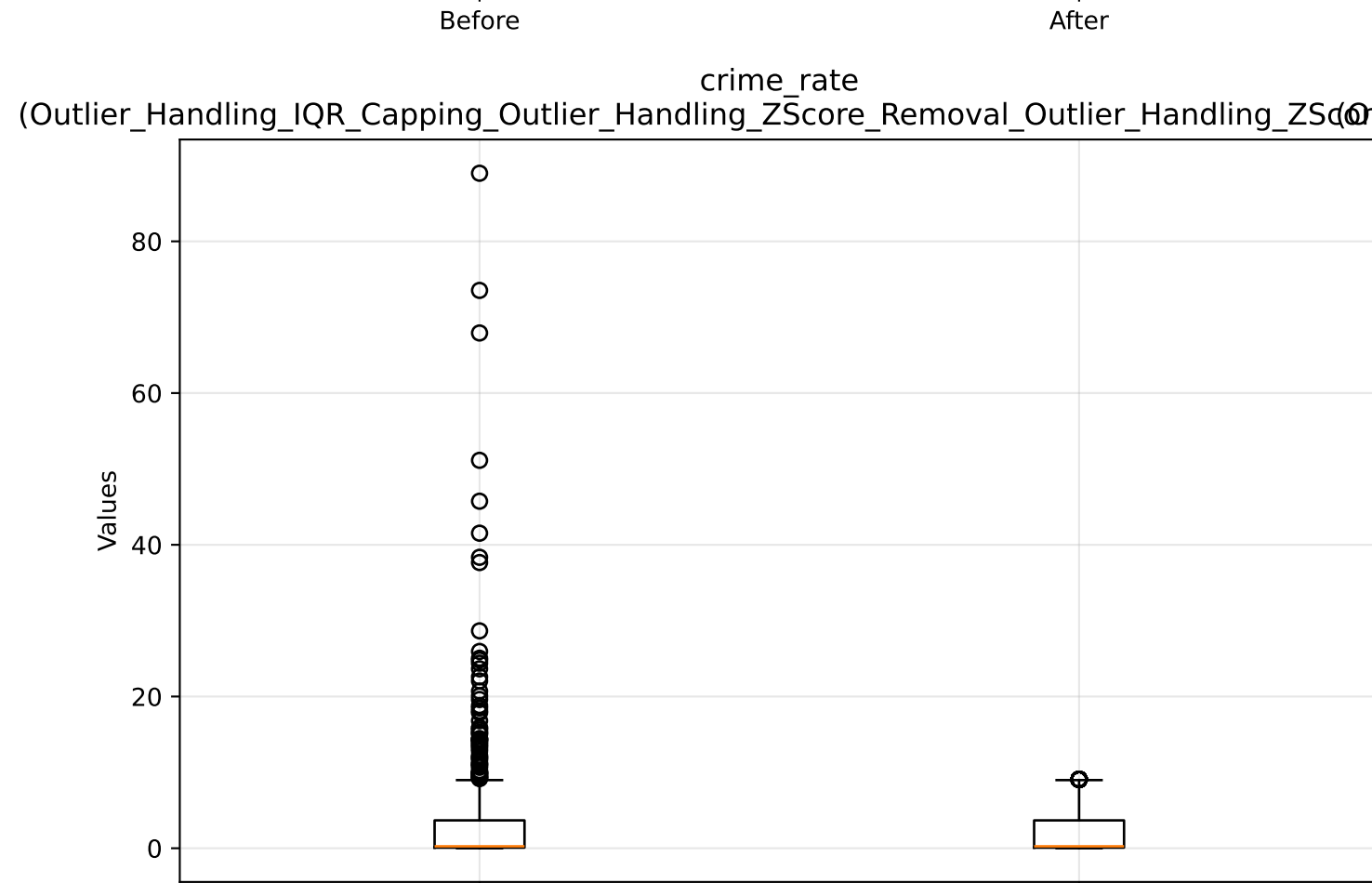
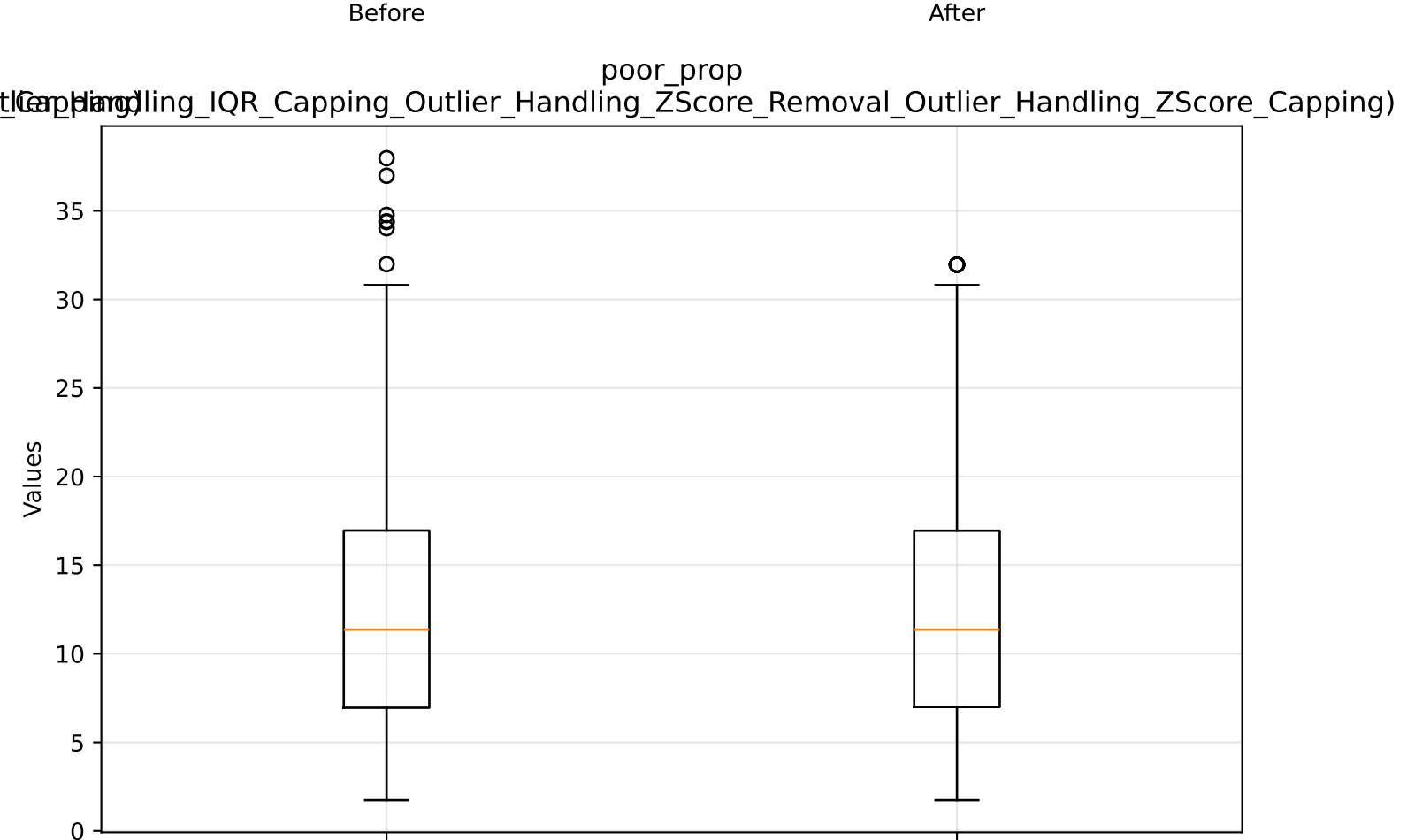
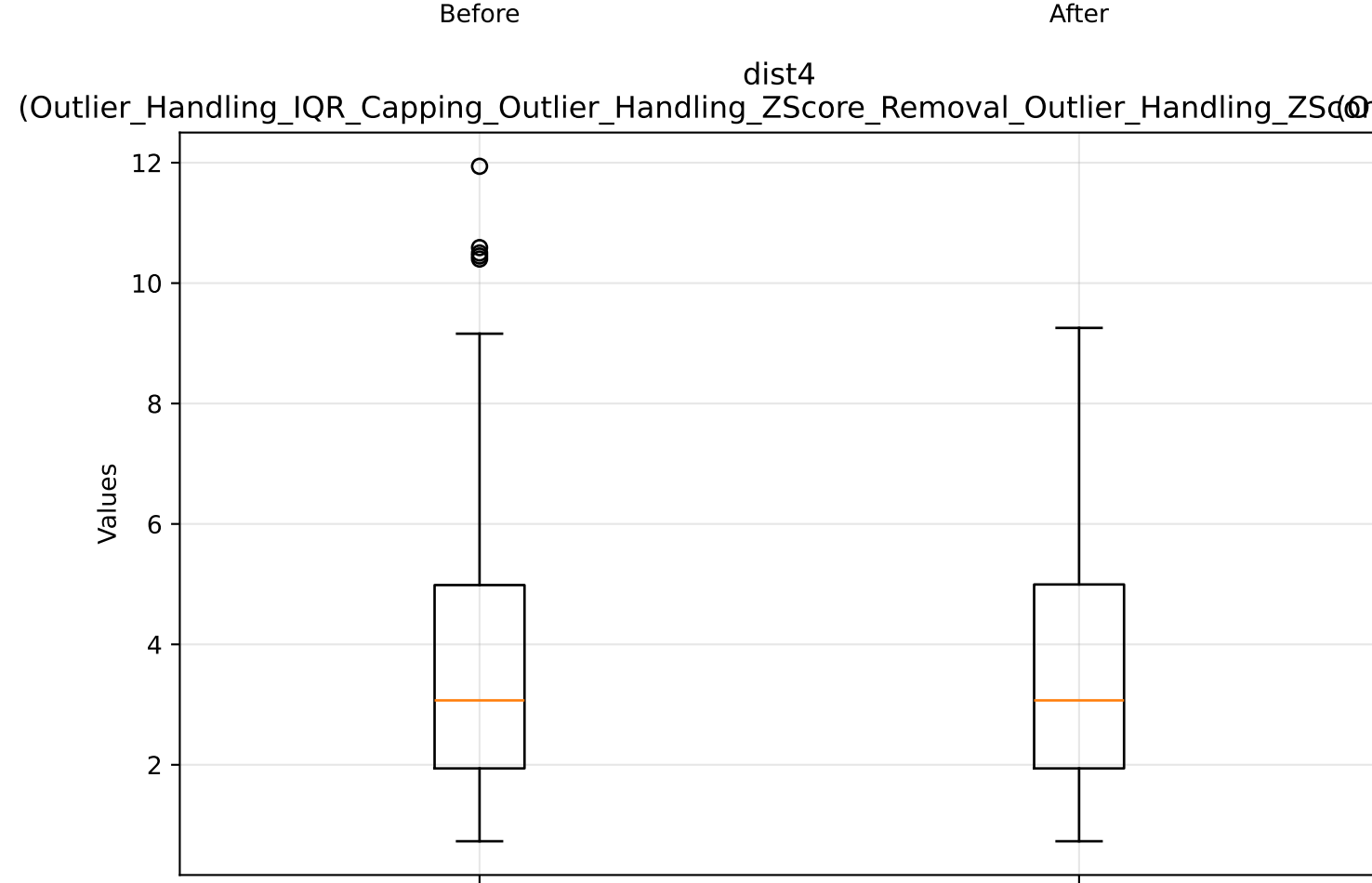
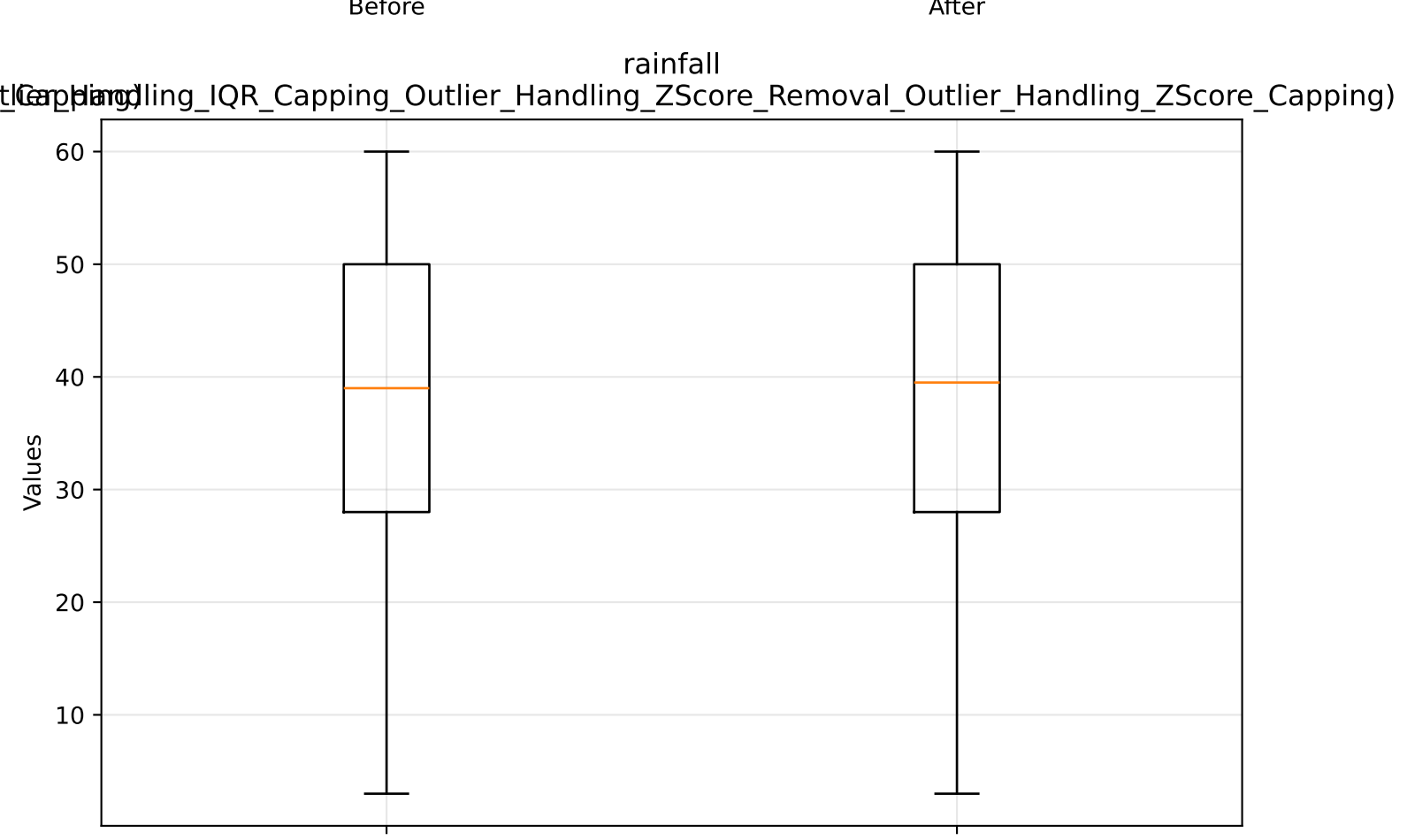
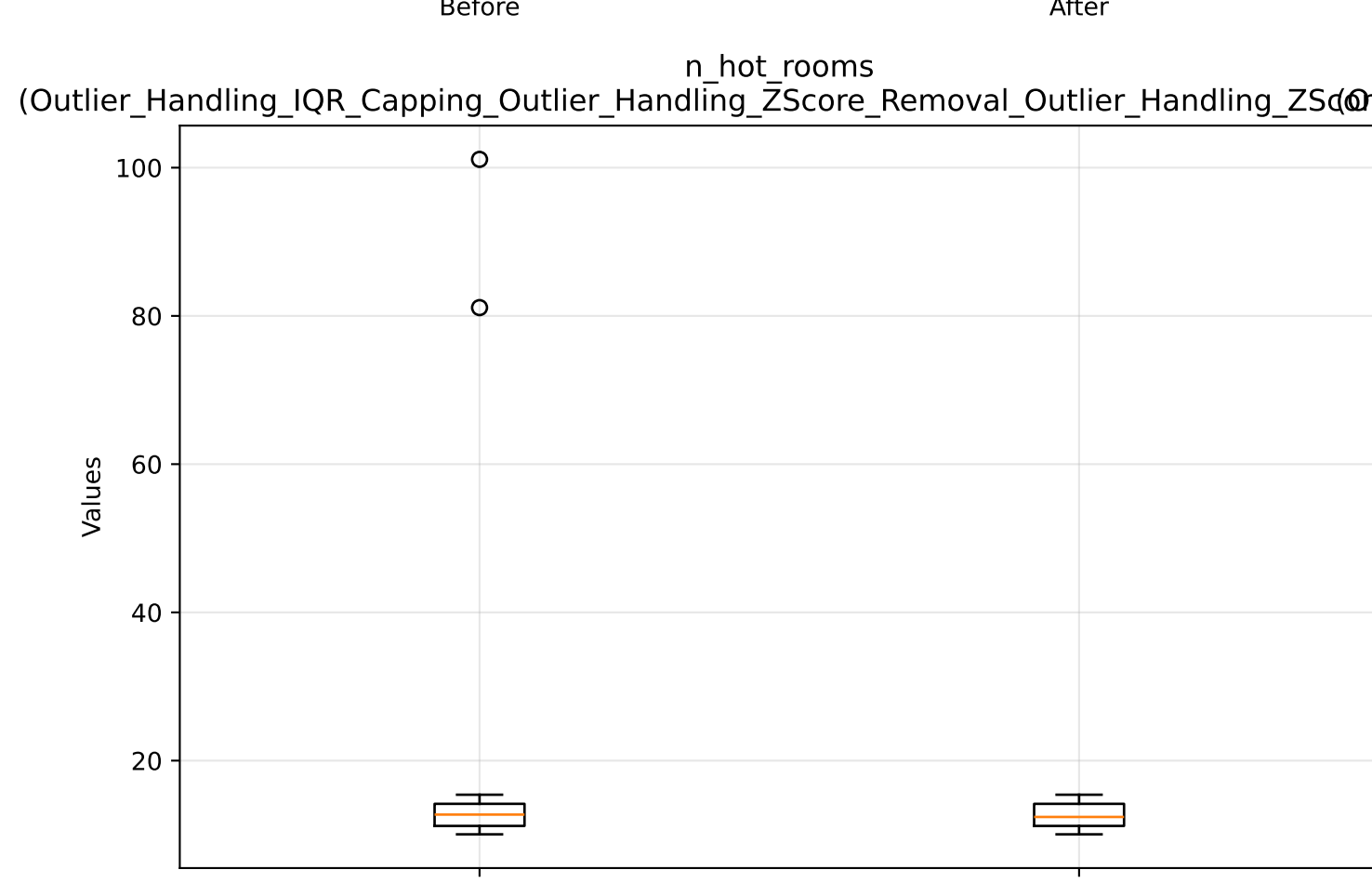
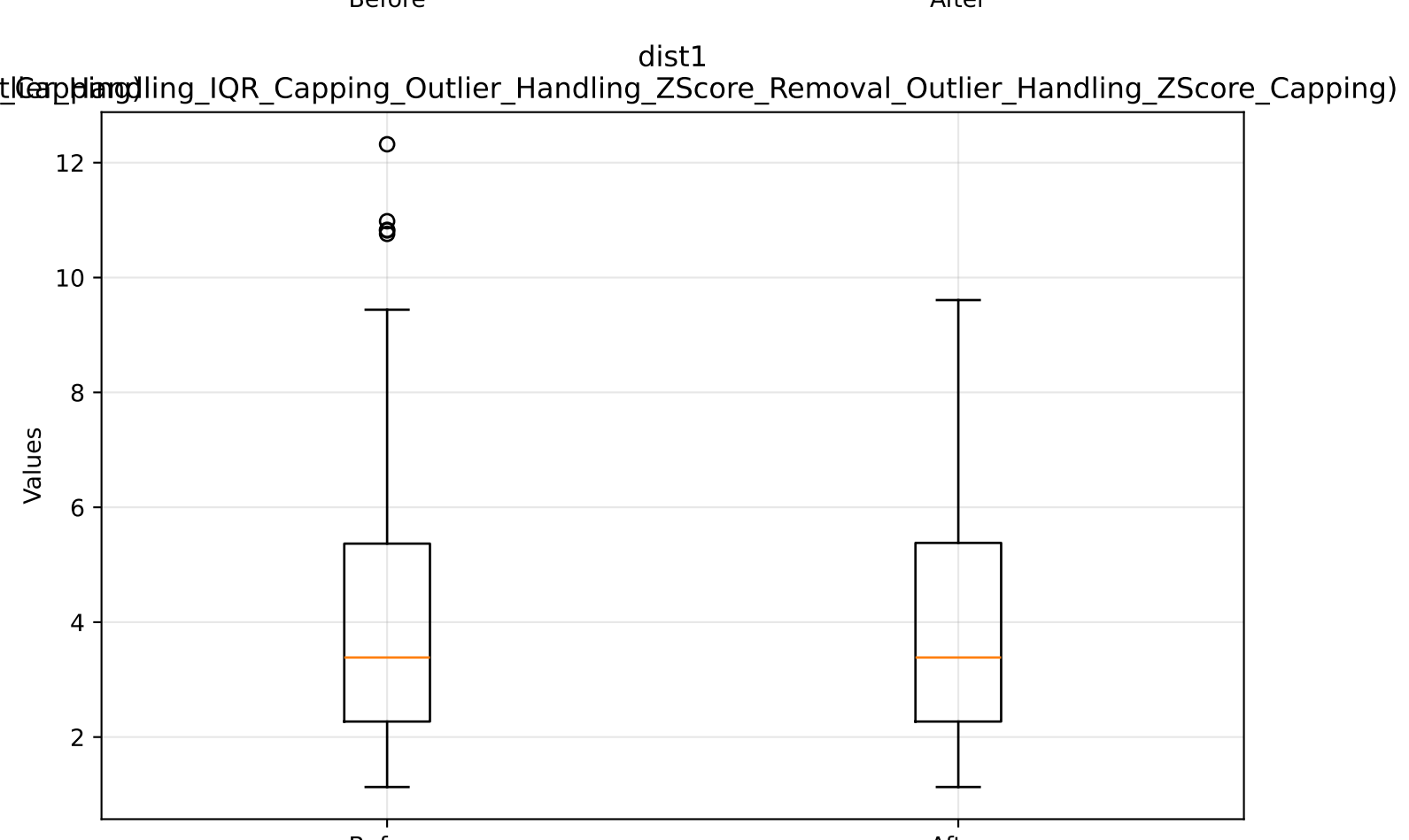
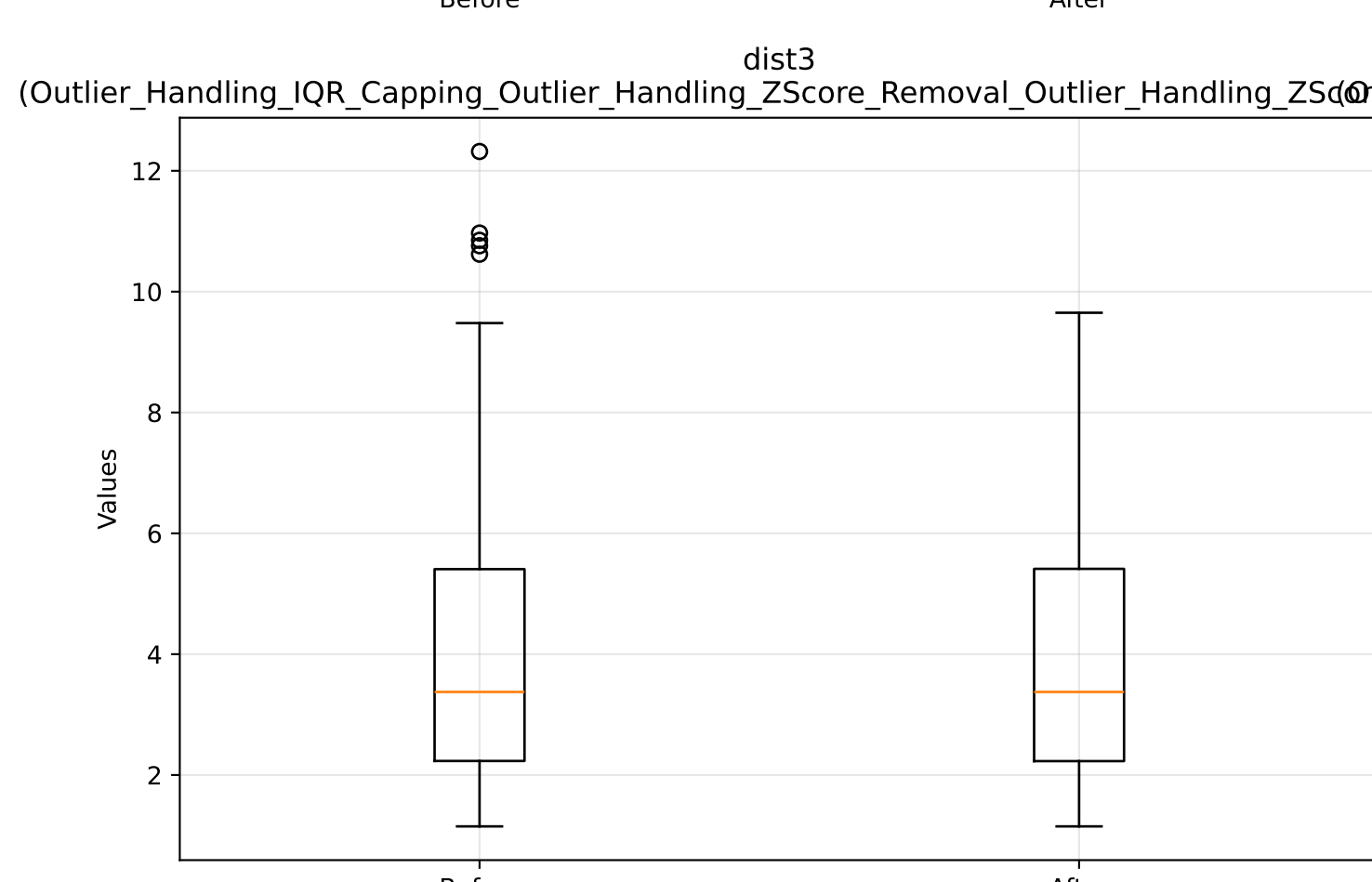
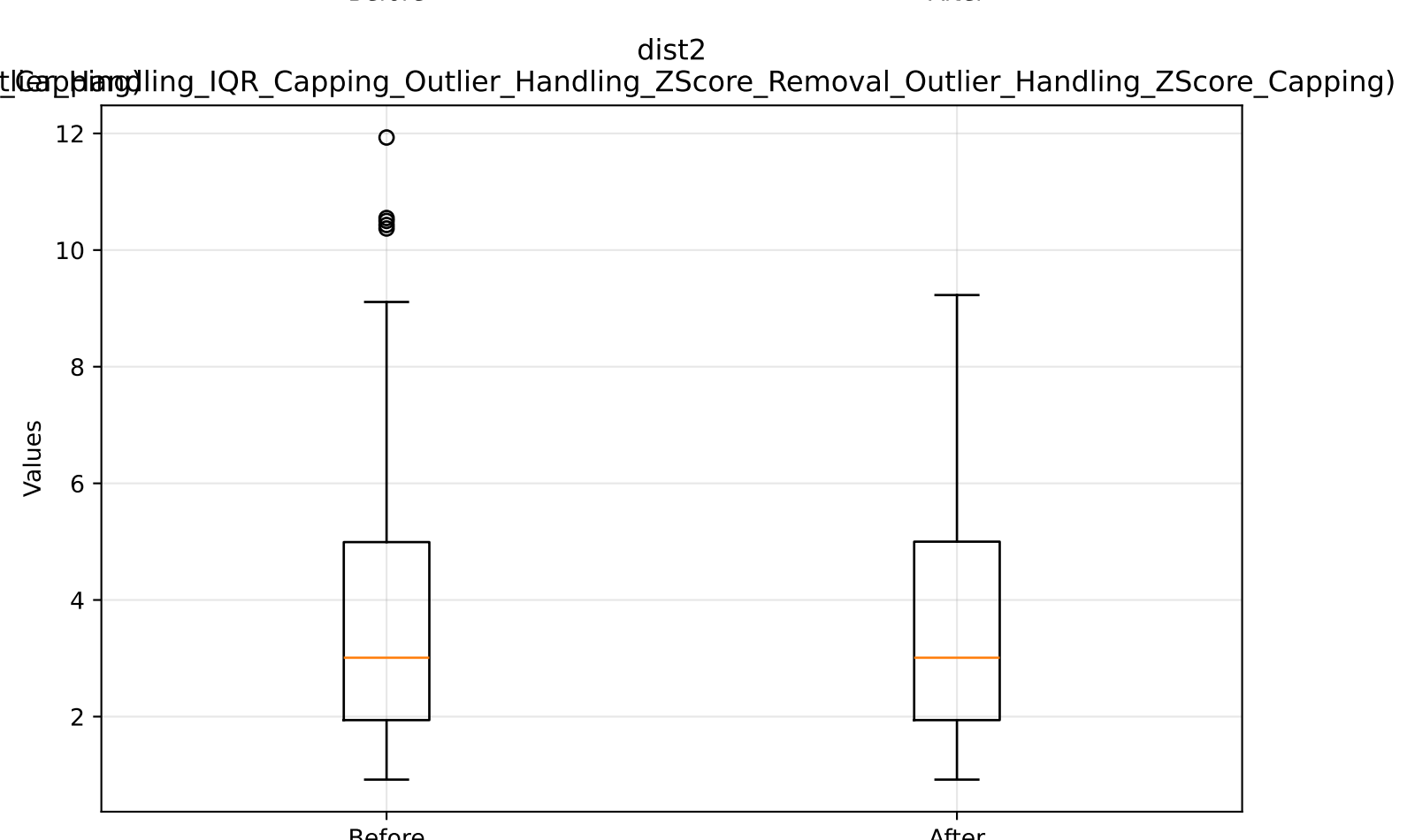
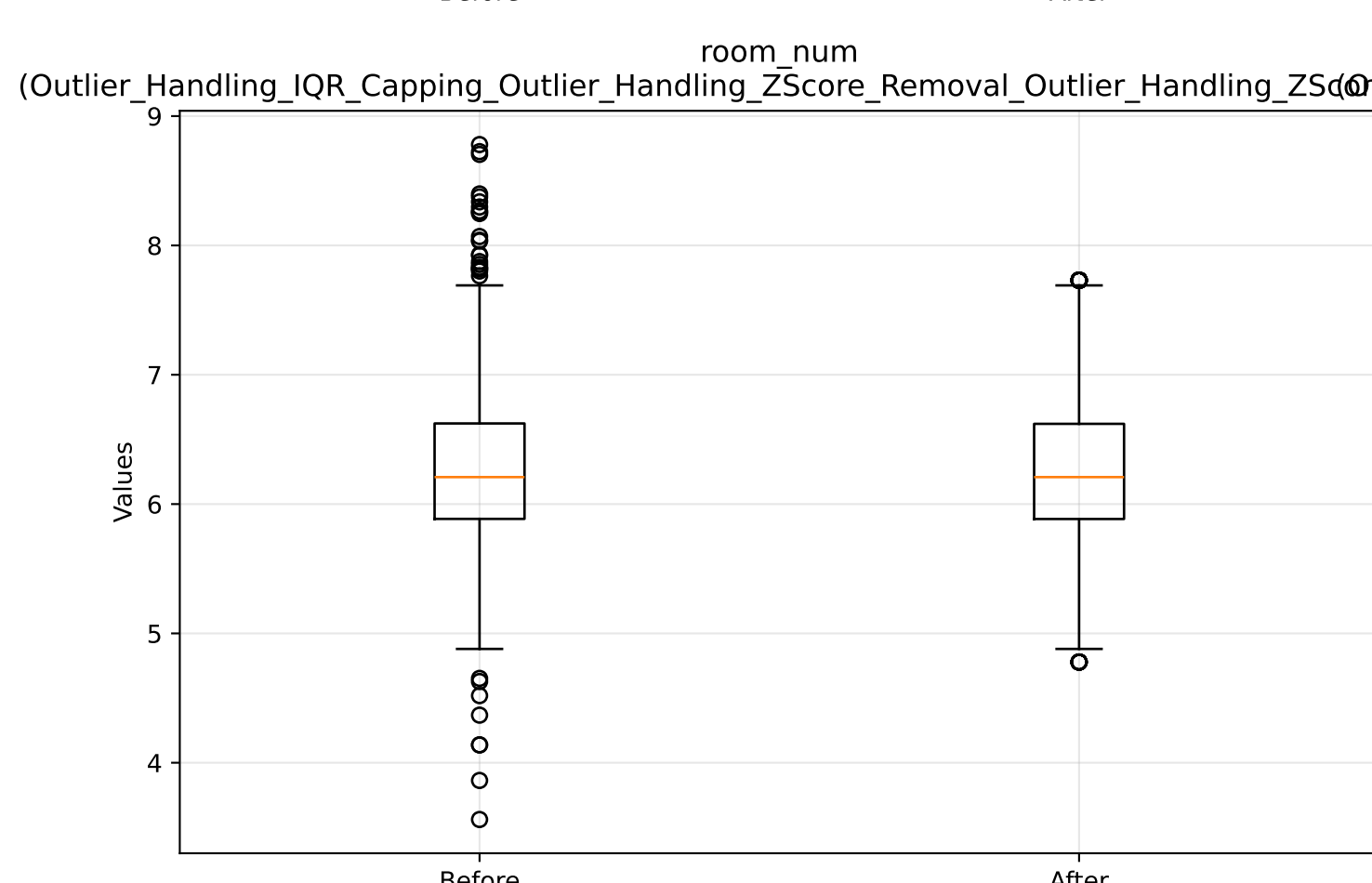
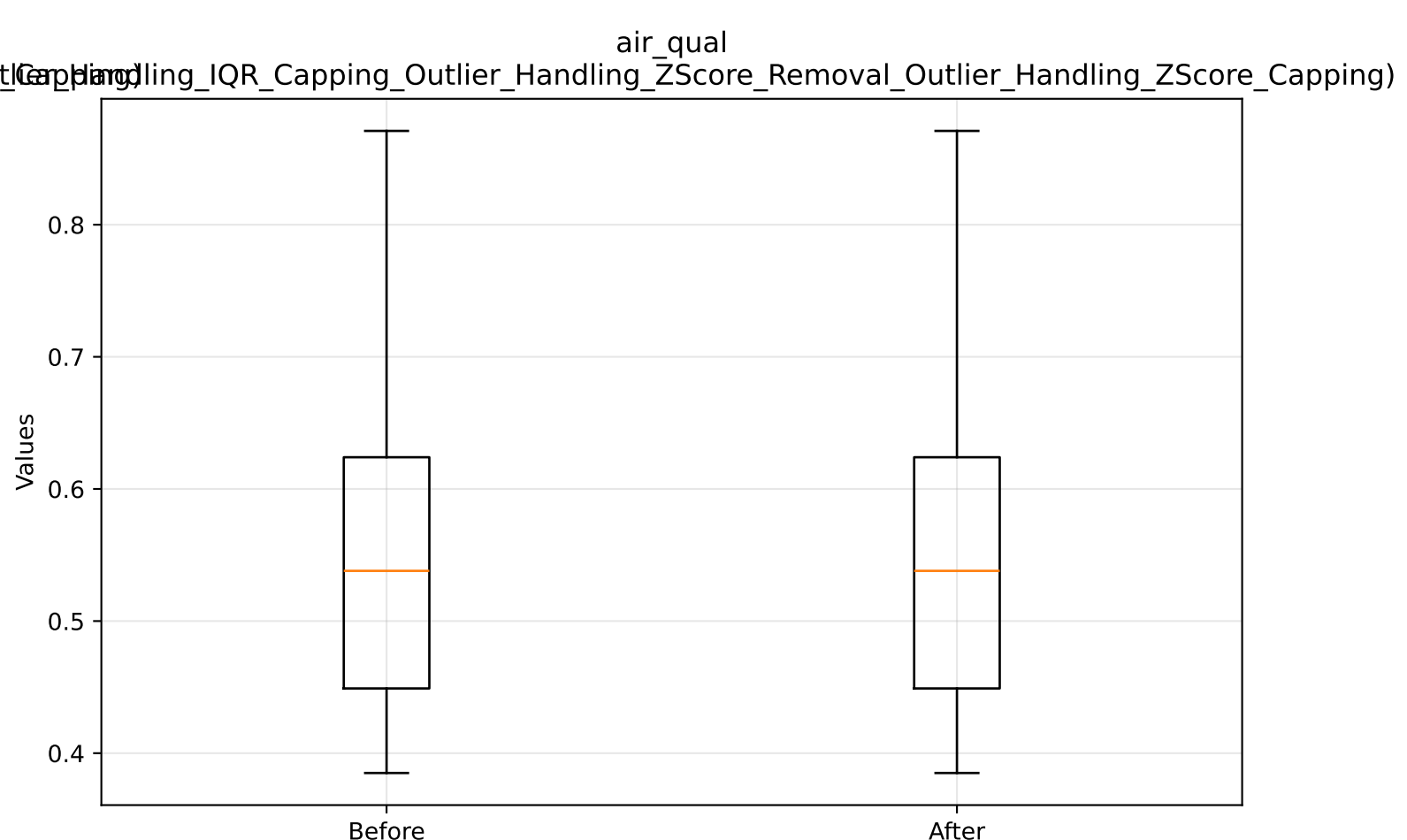
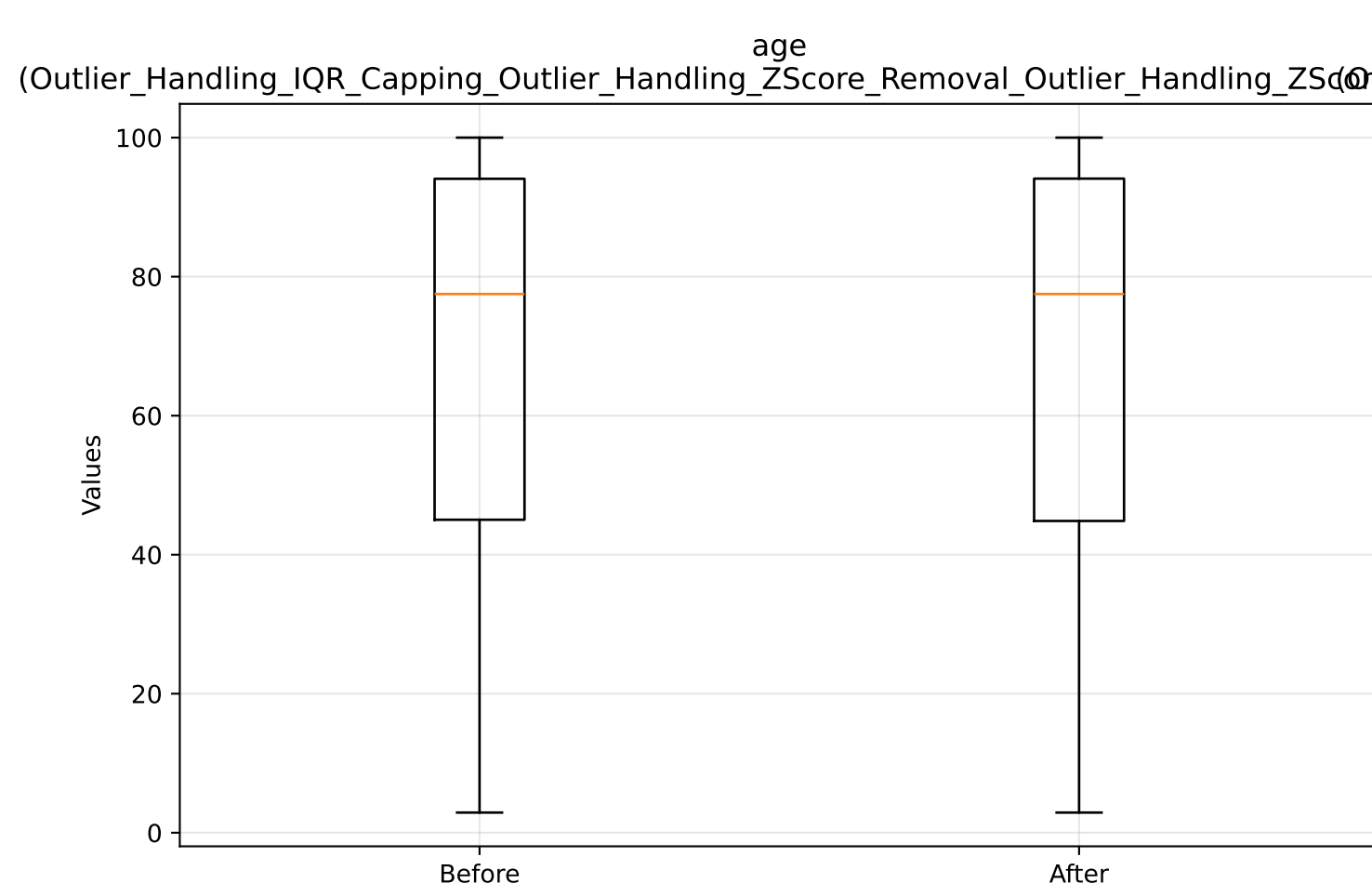
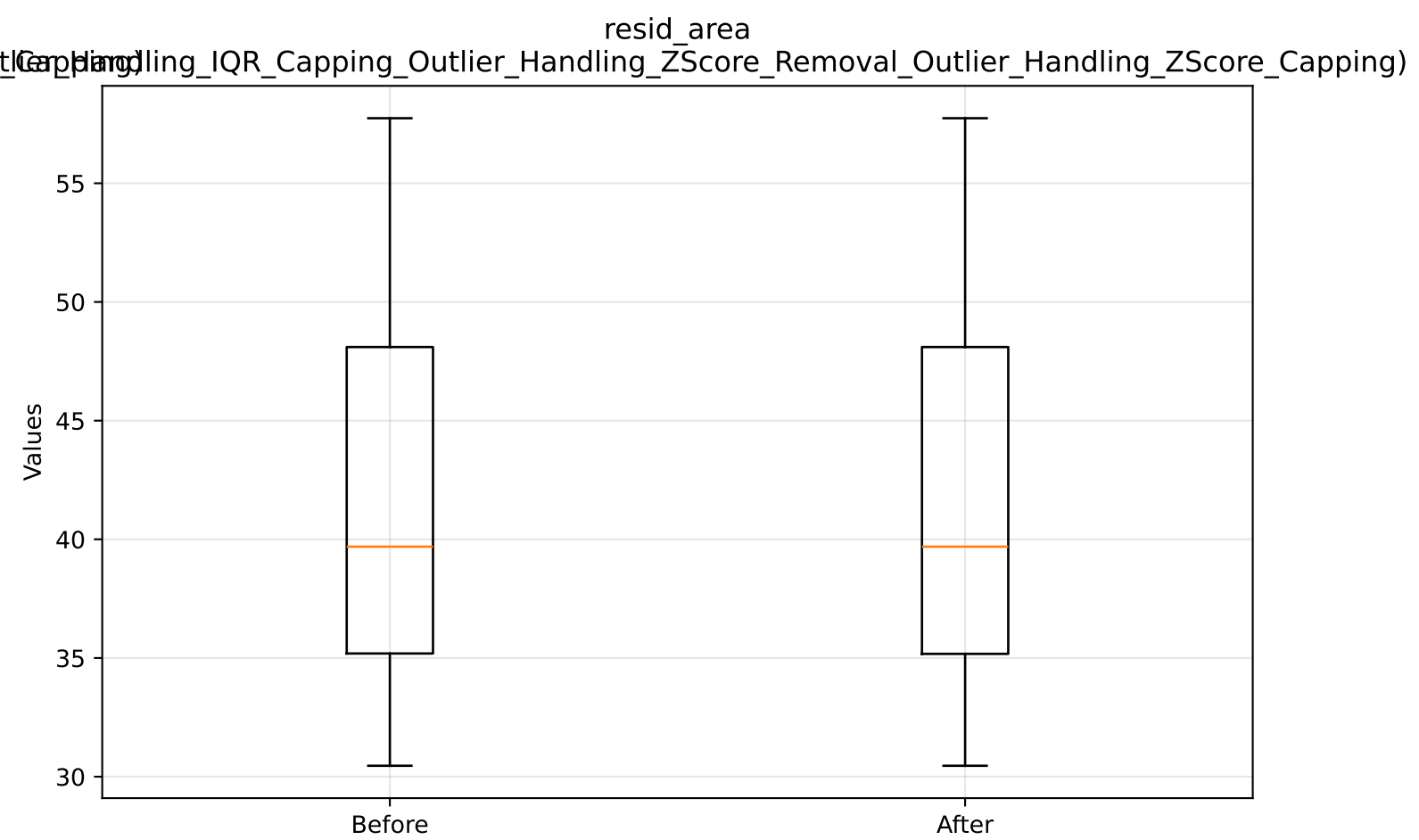
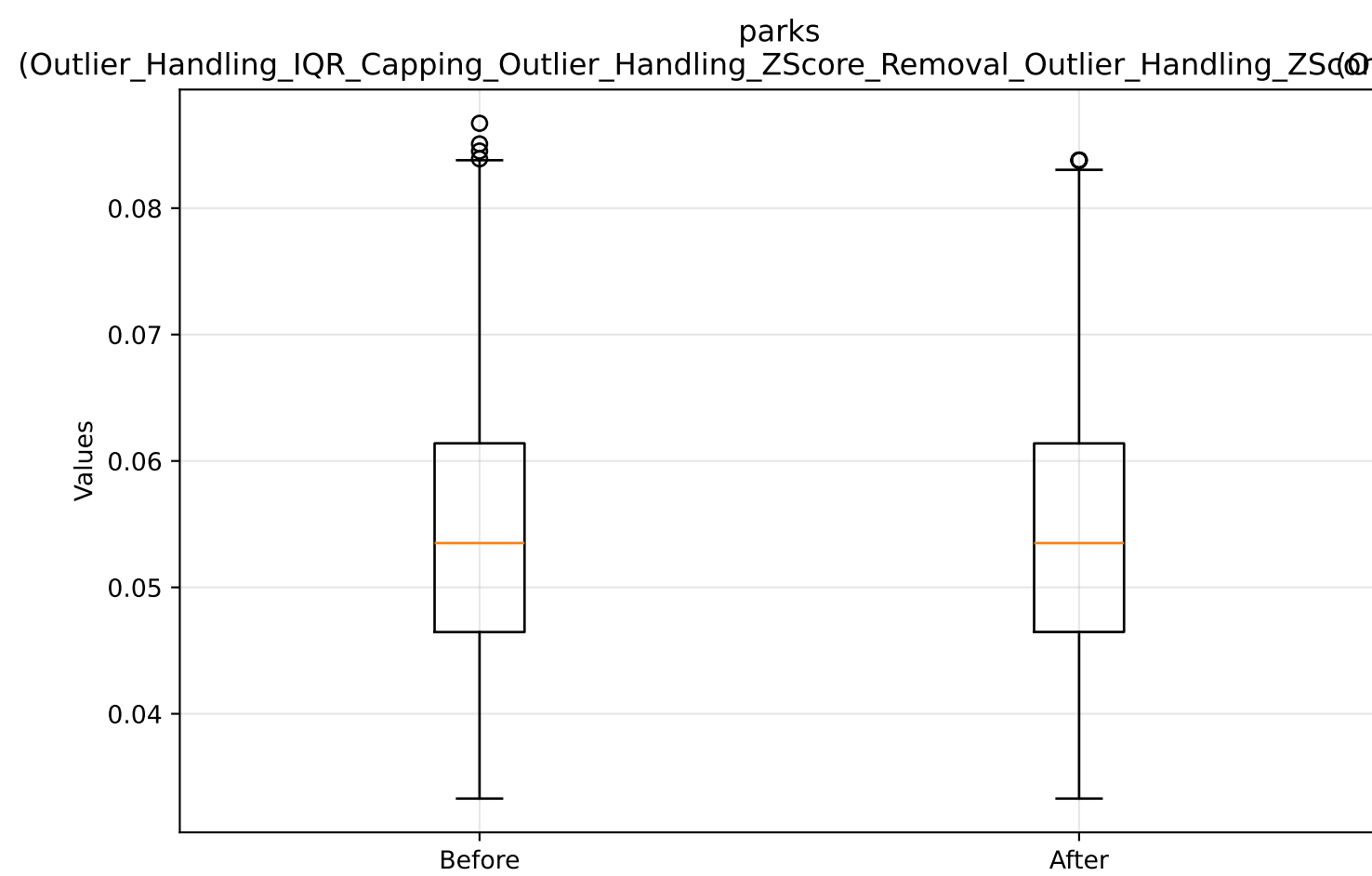
- Outlier_Handling
- Missing_Values
- Categorical_Encoding
- Dummy_Variables
- Duplicate_Removal
- Feature_Scaling
- Inconsistent_Data

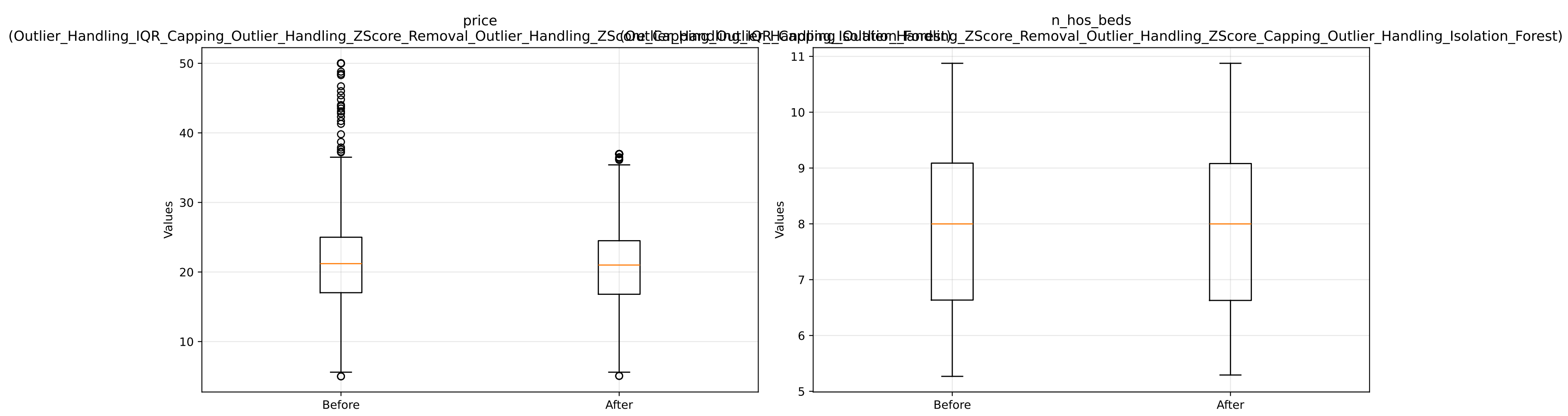
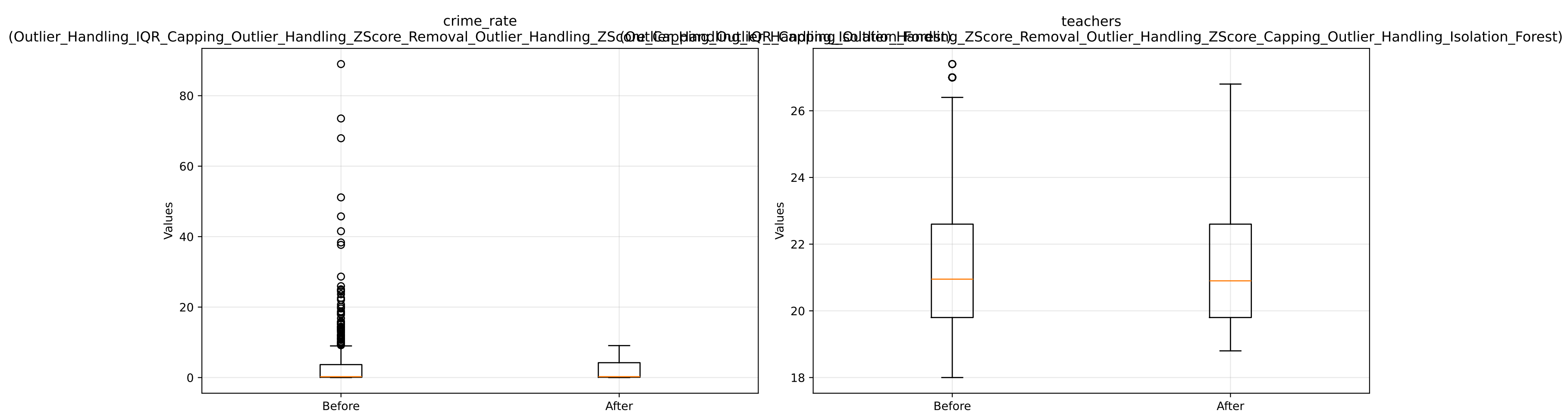
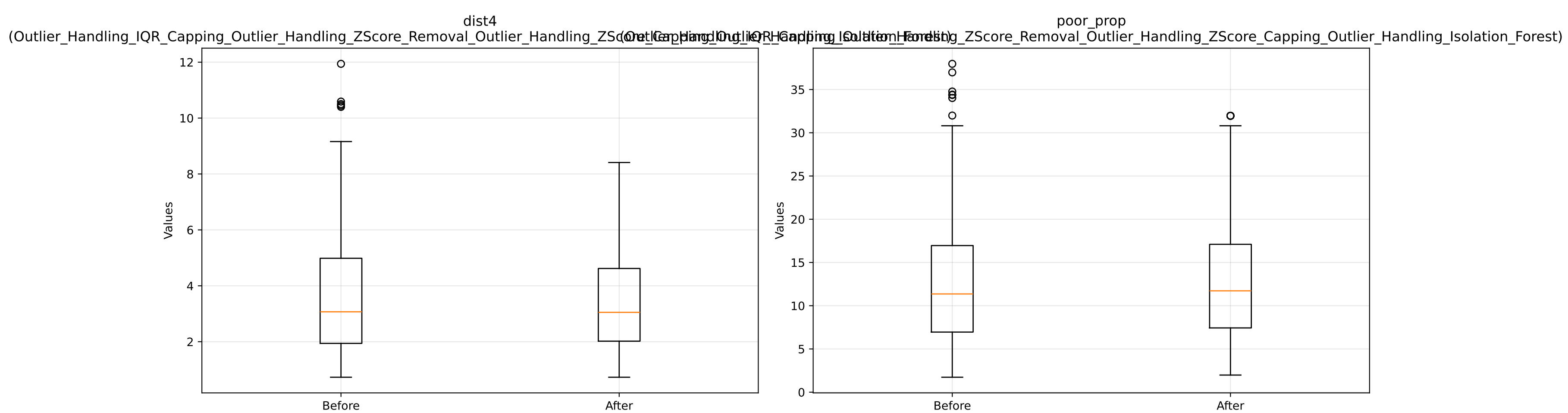
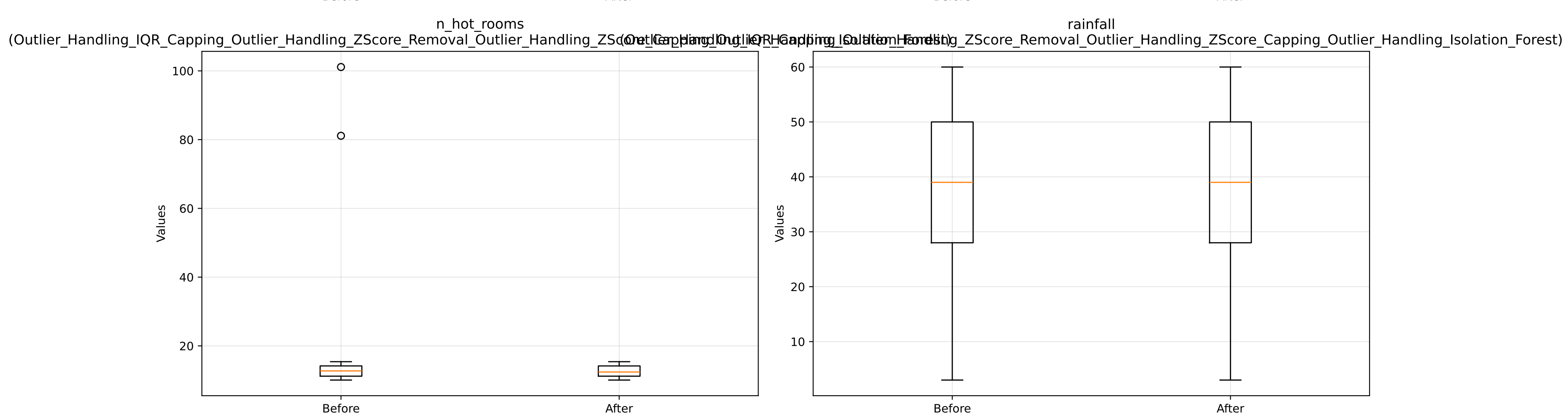
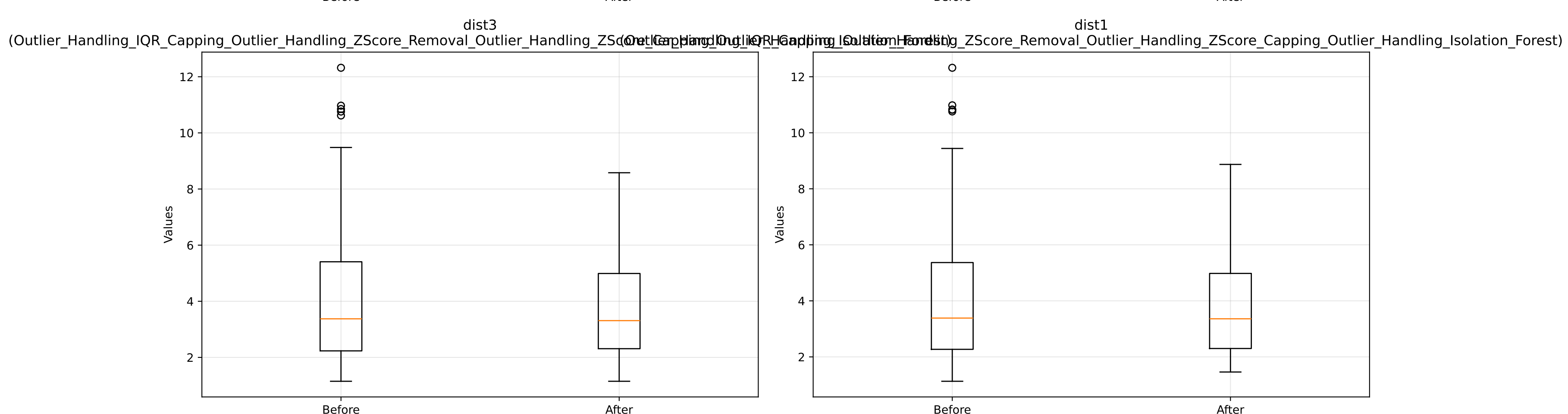
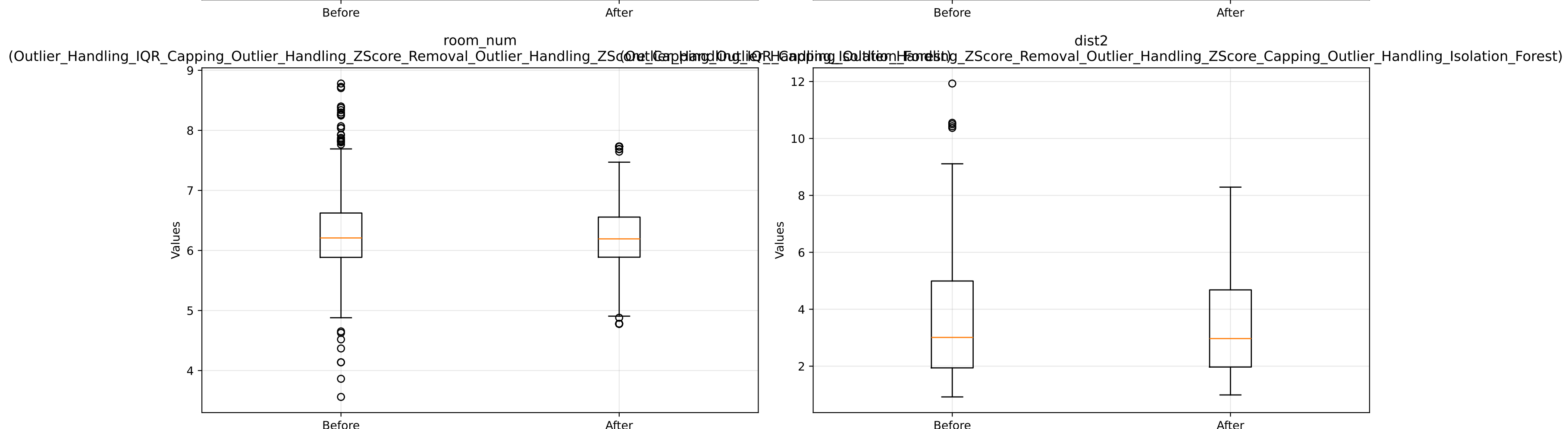
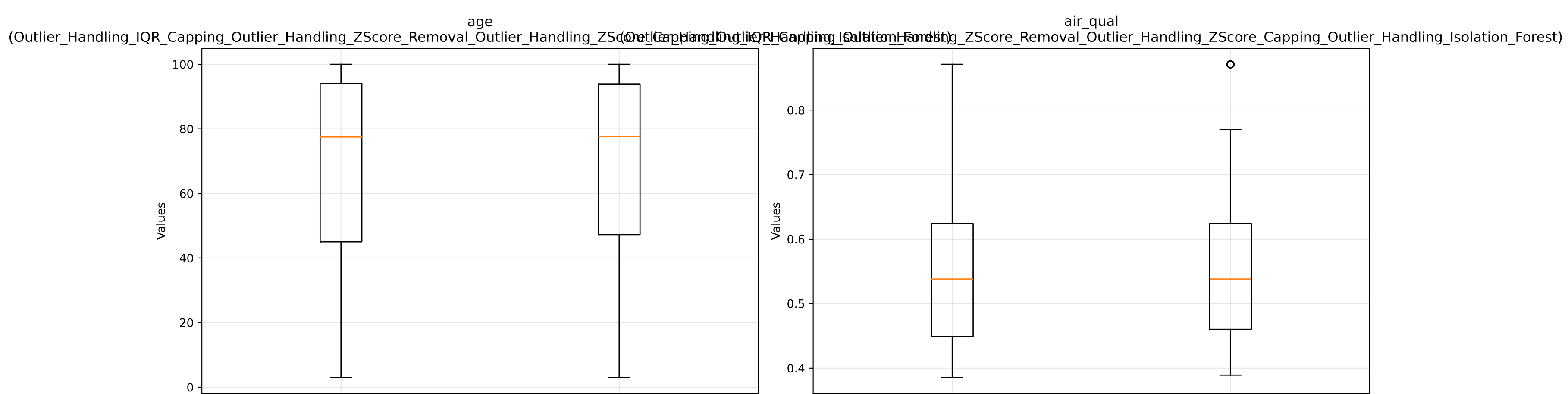
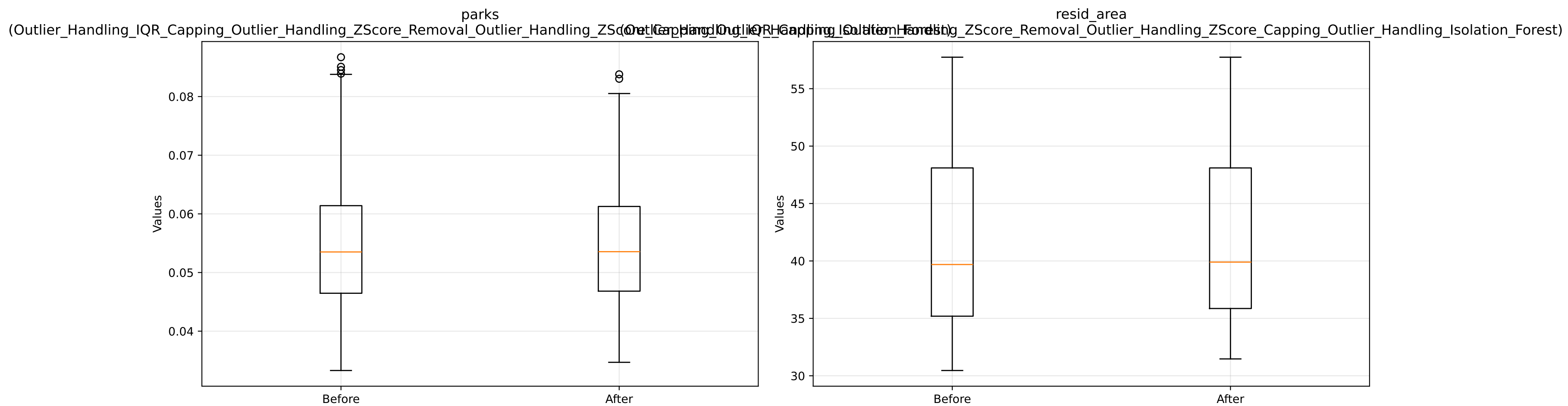
Output Files:

- CSV files for each processing combination
- PDF processing report

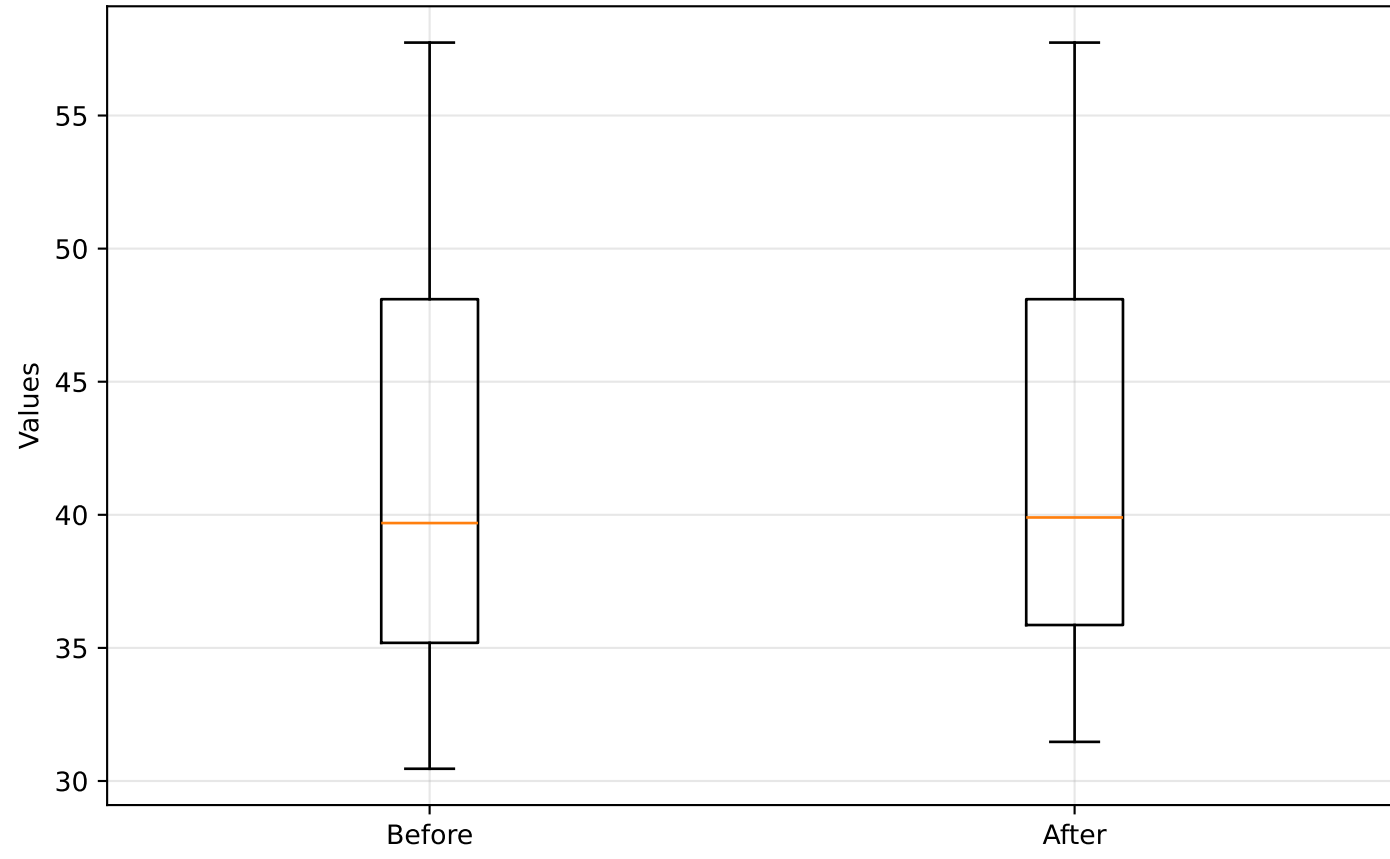
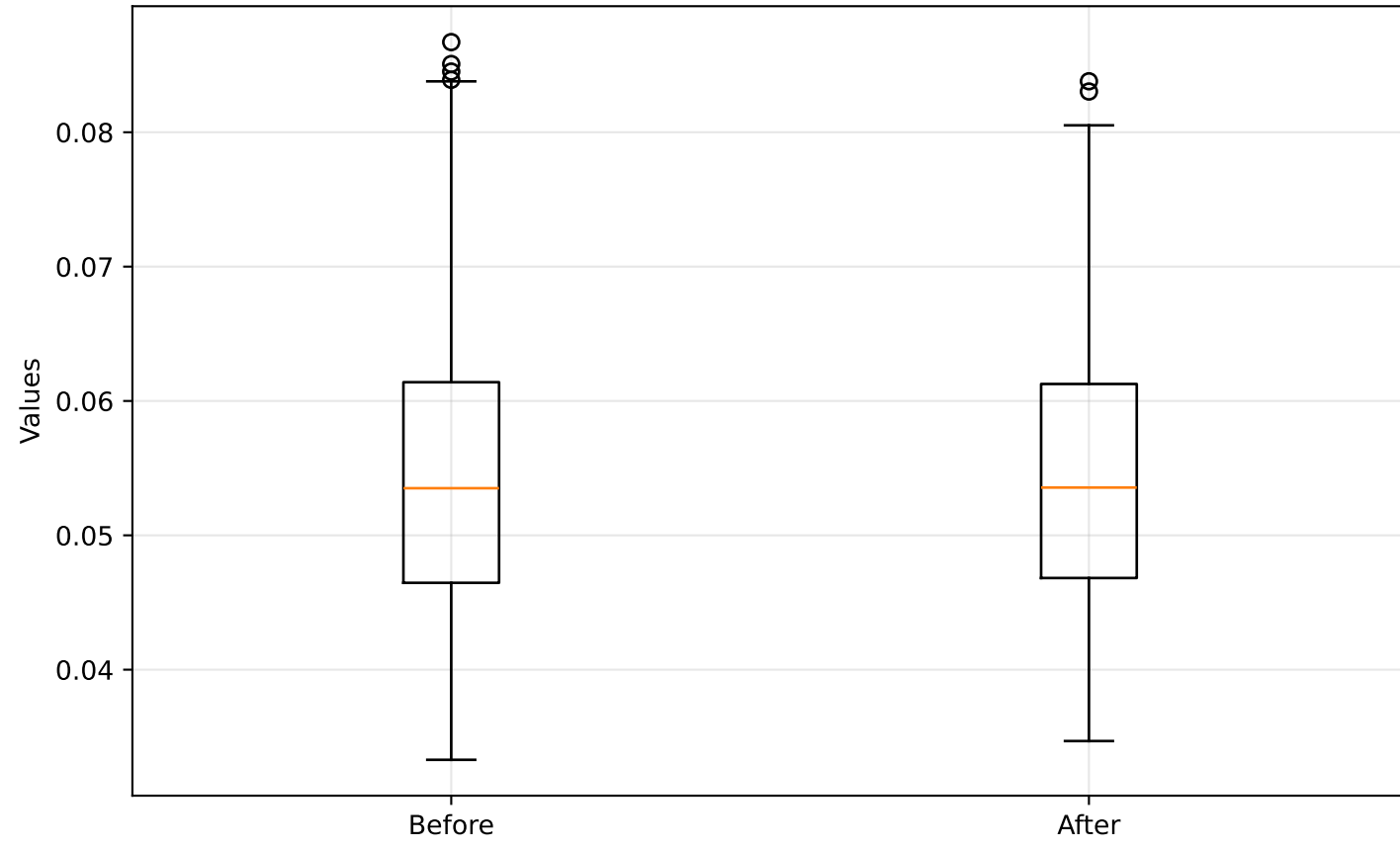




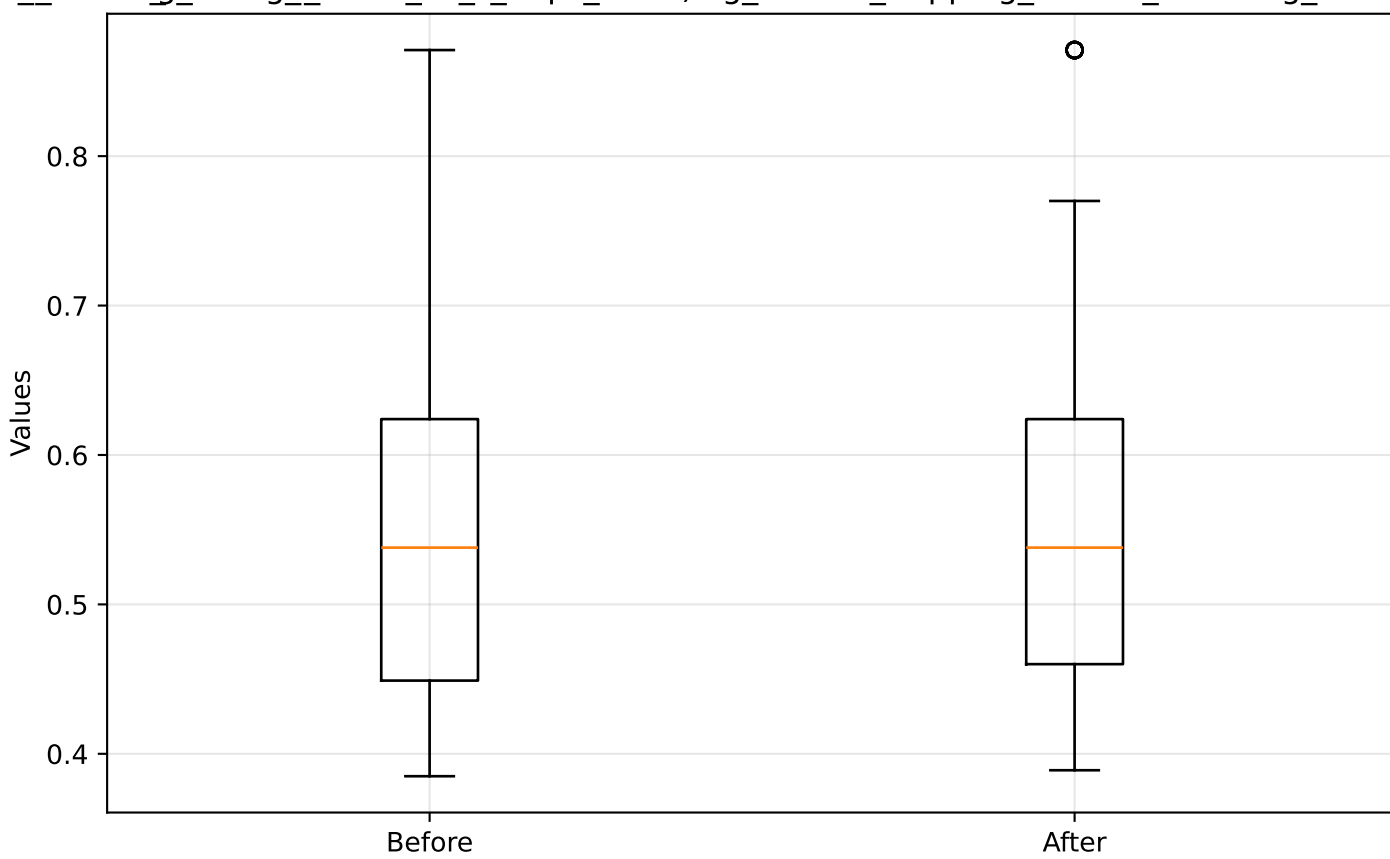
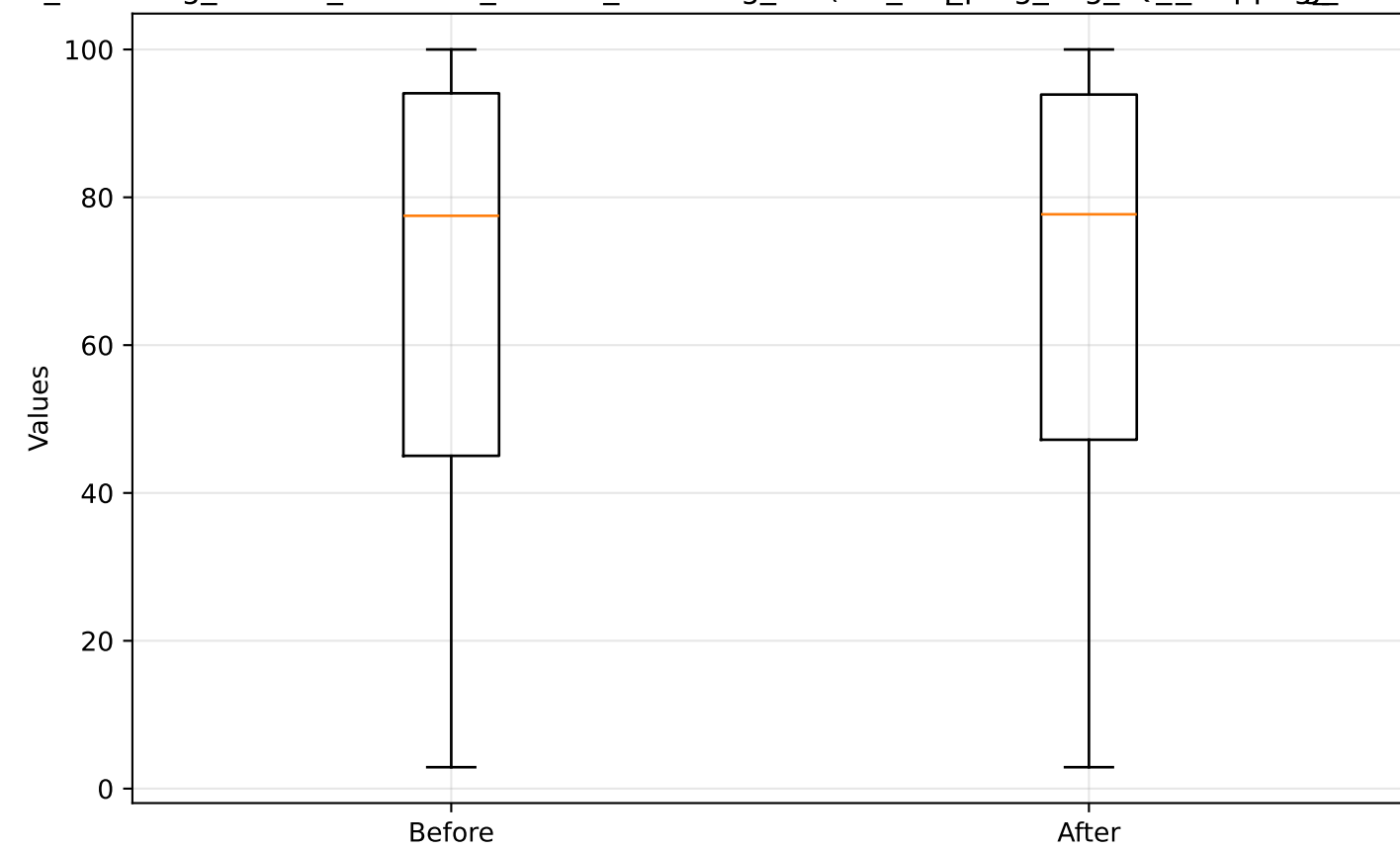




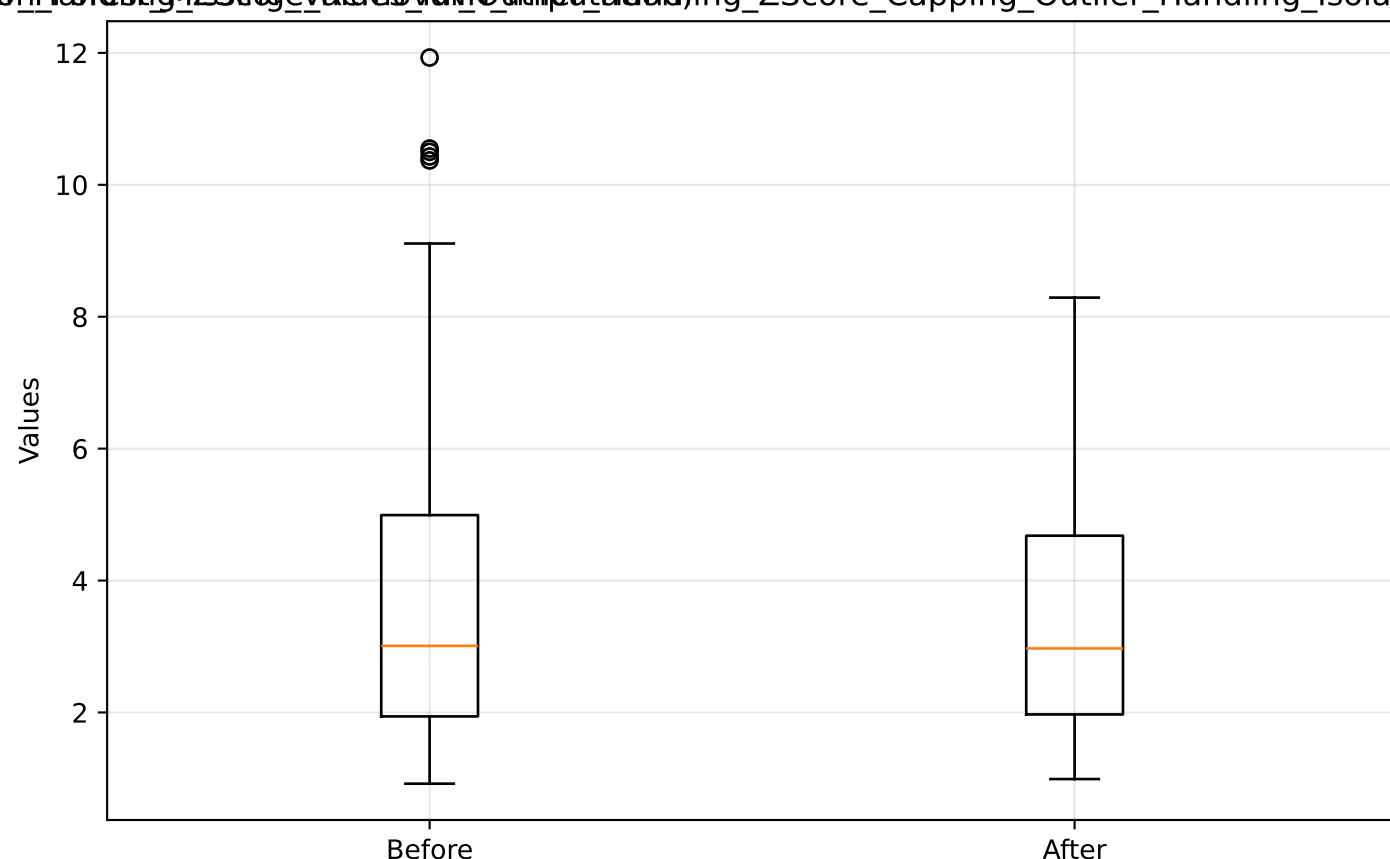
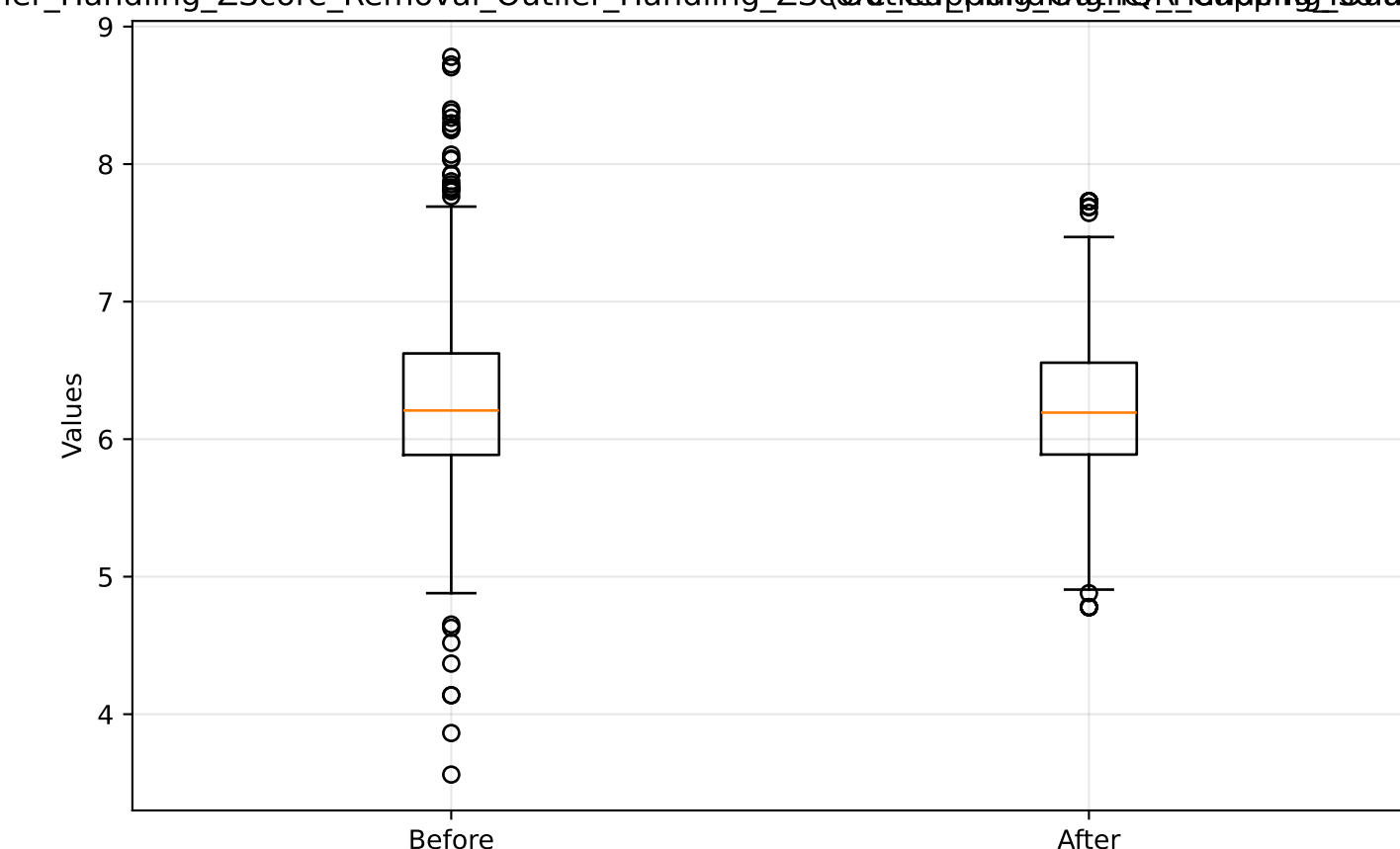
(Outlier_Handling_IQR_Capping_Outlier_Handling_ZScore_Removal_Outlier_Handling_ZScore_Capping_Outlier_Handling_Isolation_Forest_Missing_Value_KNN_Outlier_Detection)



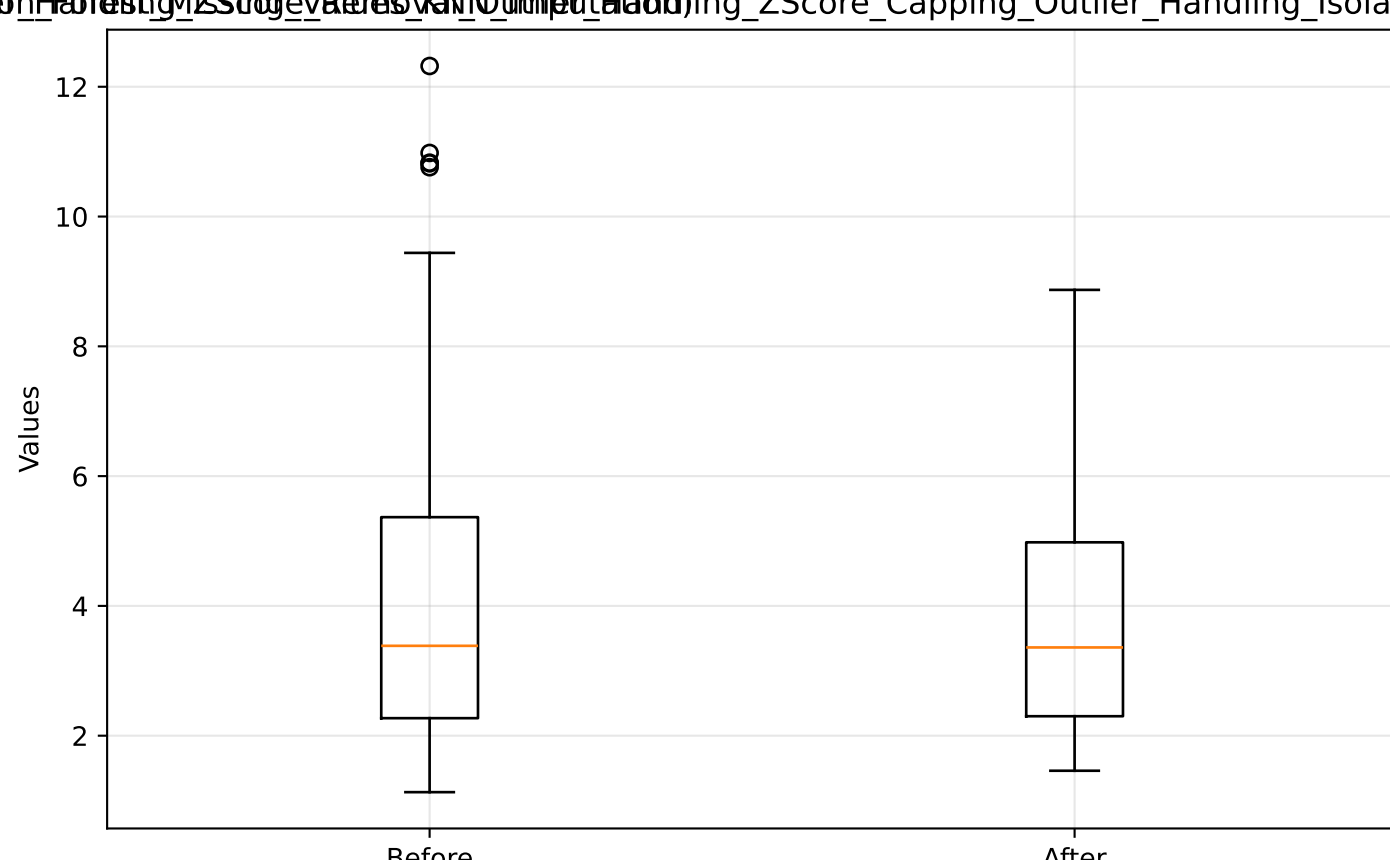
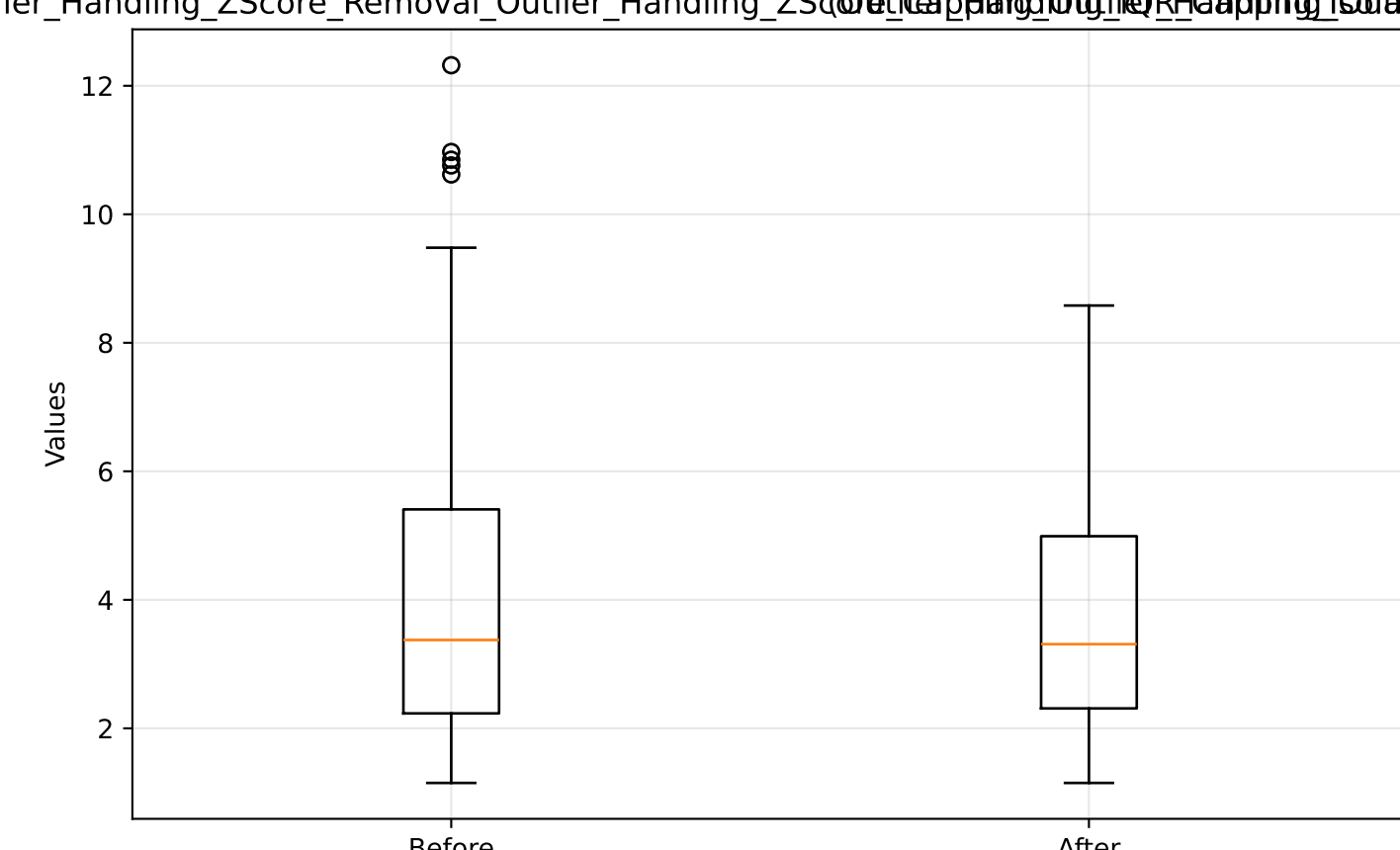
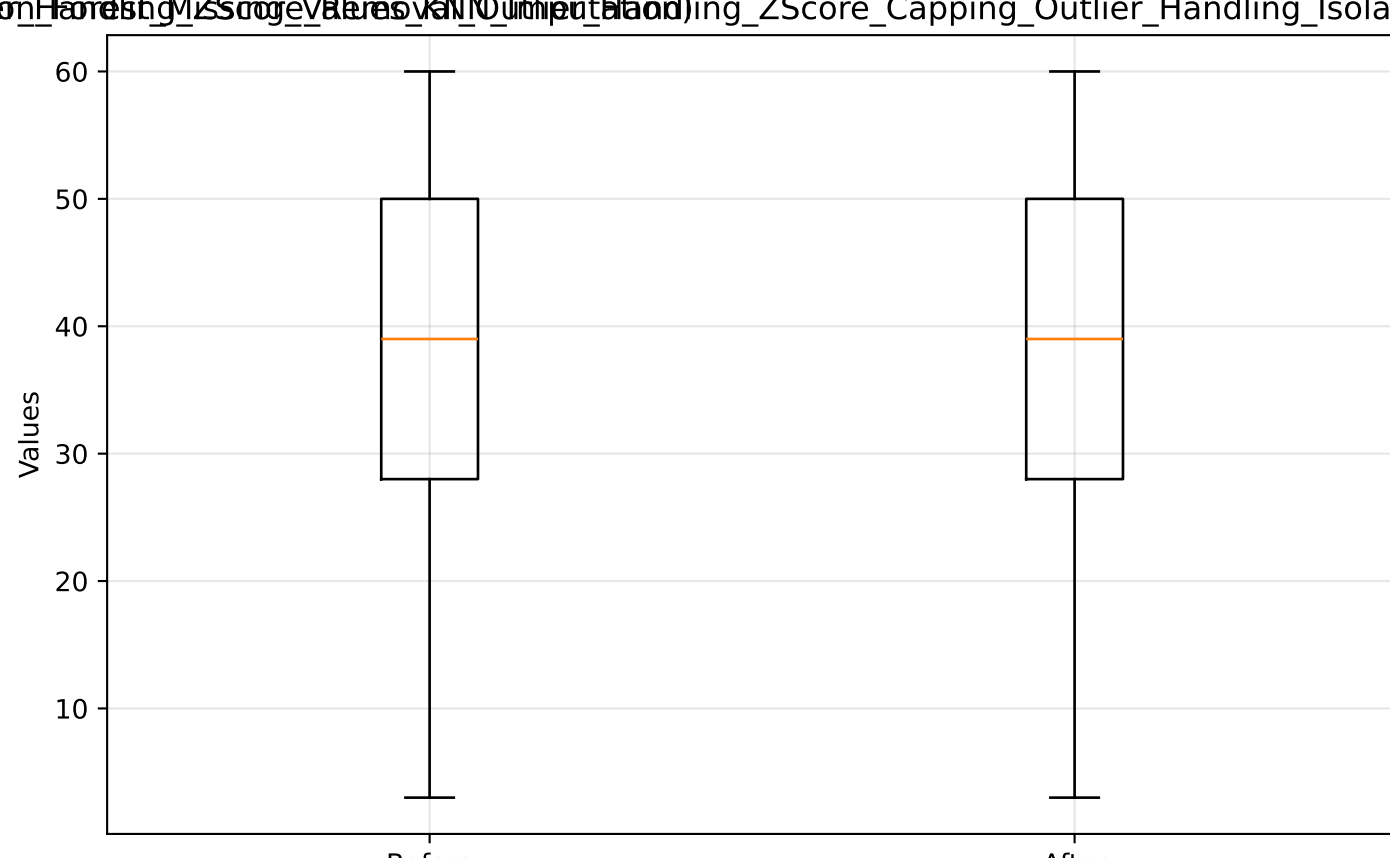
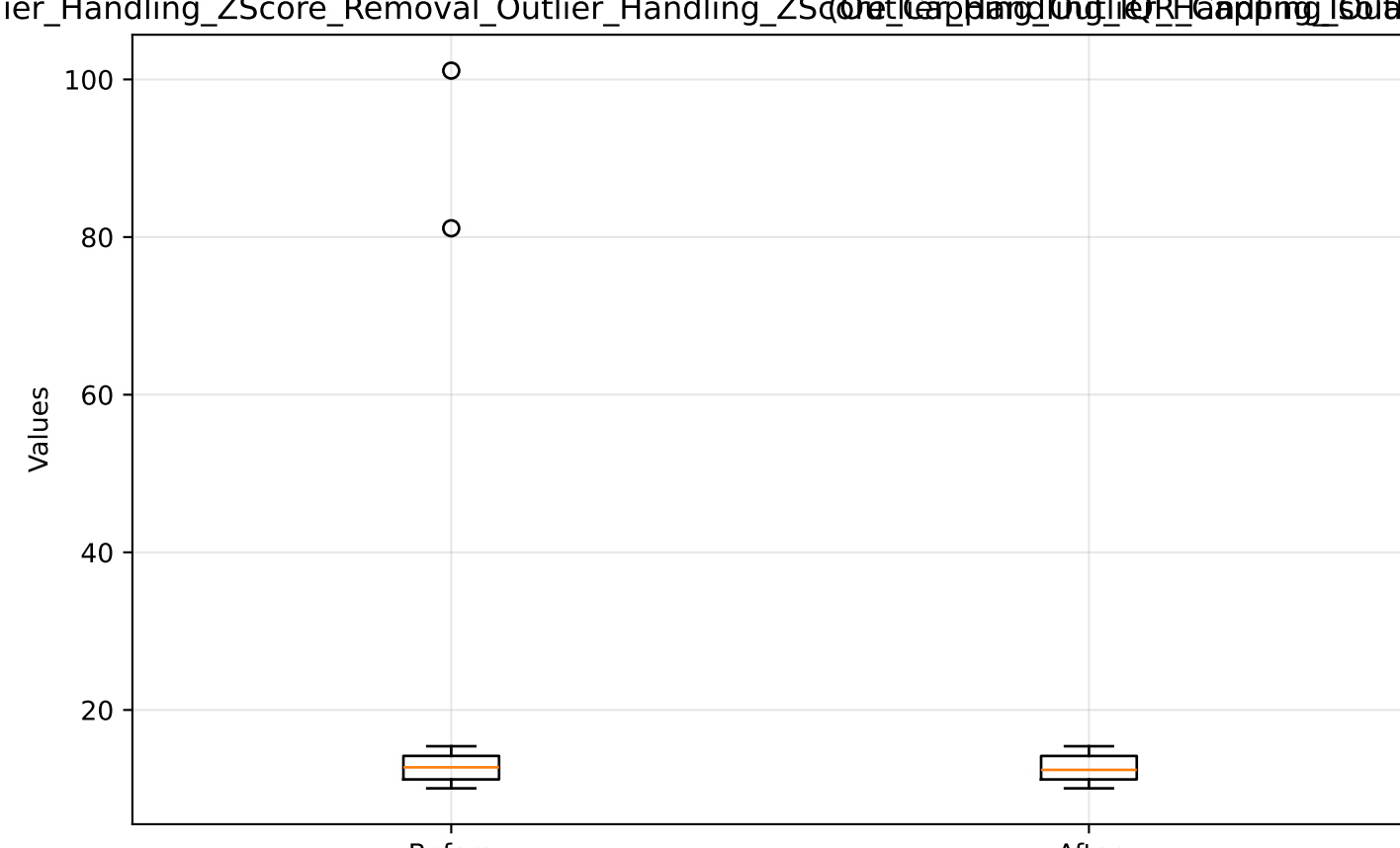
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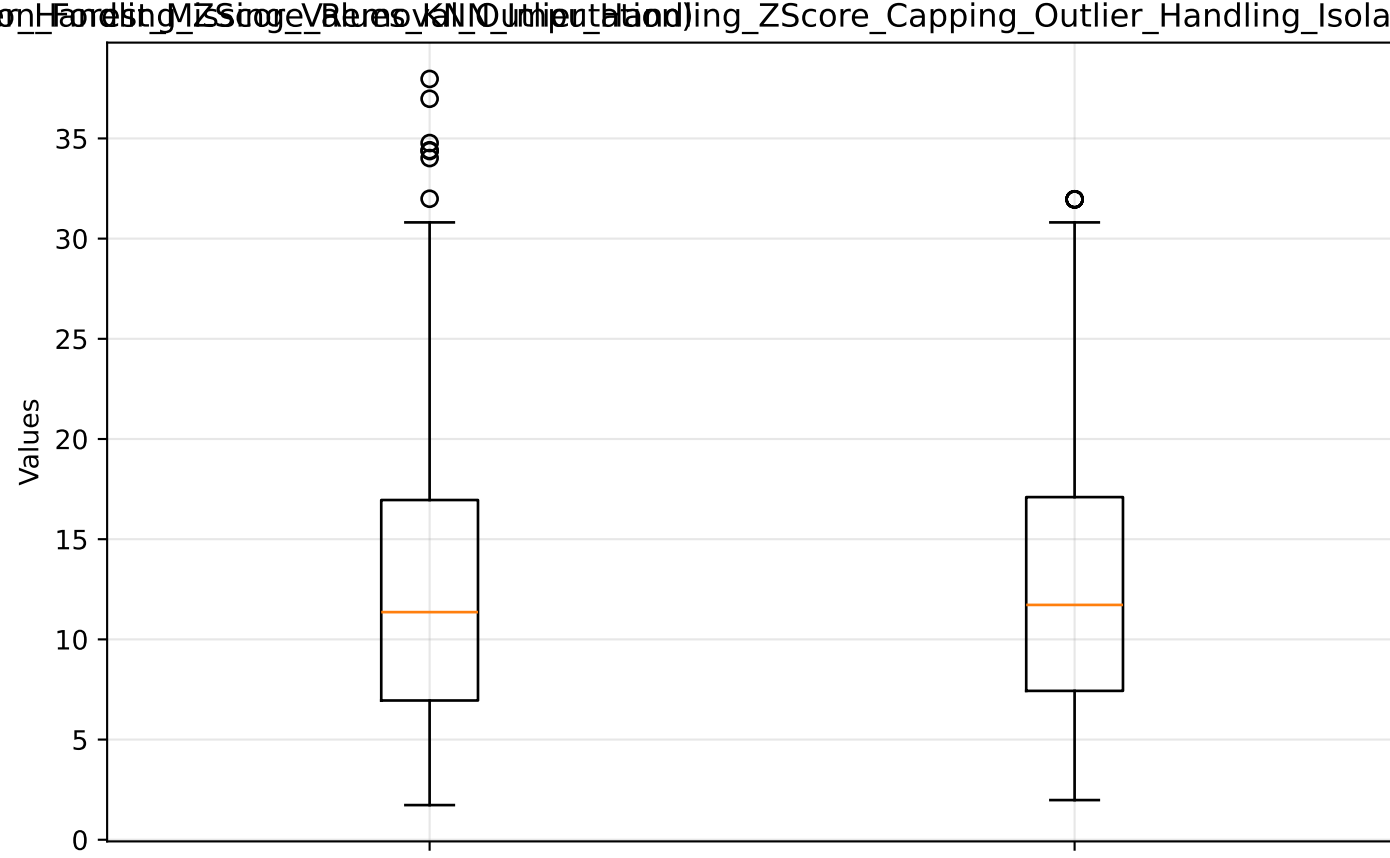
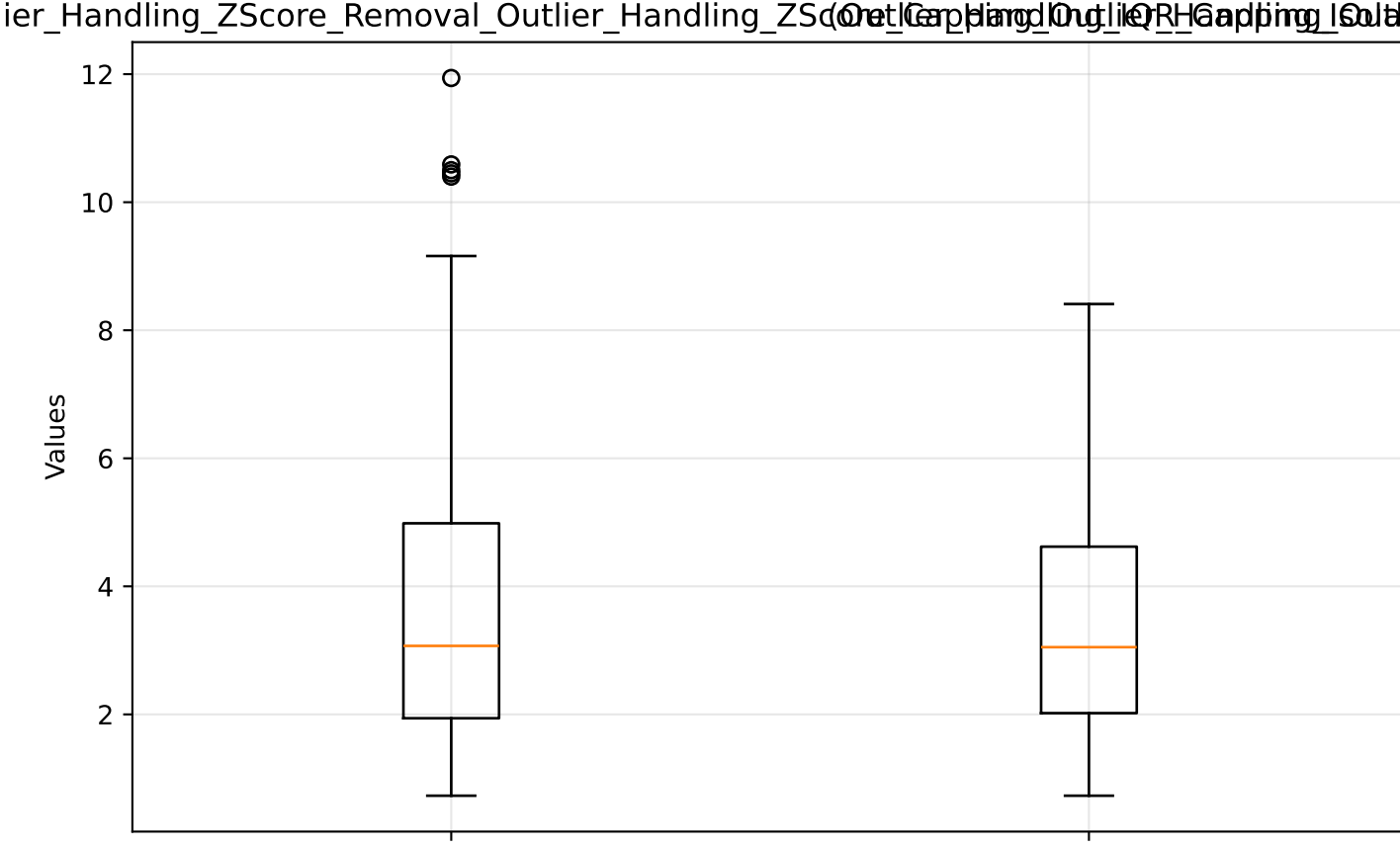
	room_num	dist2
(Outlier Handling IQR Capping Outlier Handling ZScore Removal Outlier Handling Isolation Forest Missing Value Imputation)		



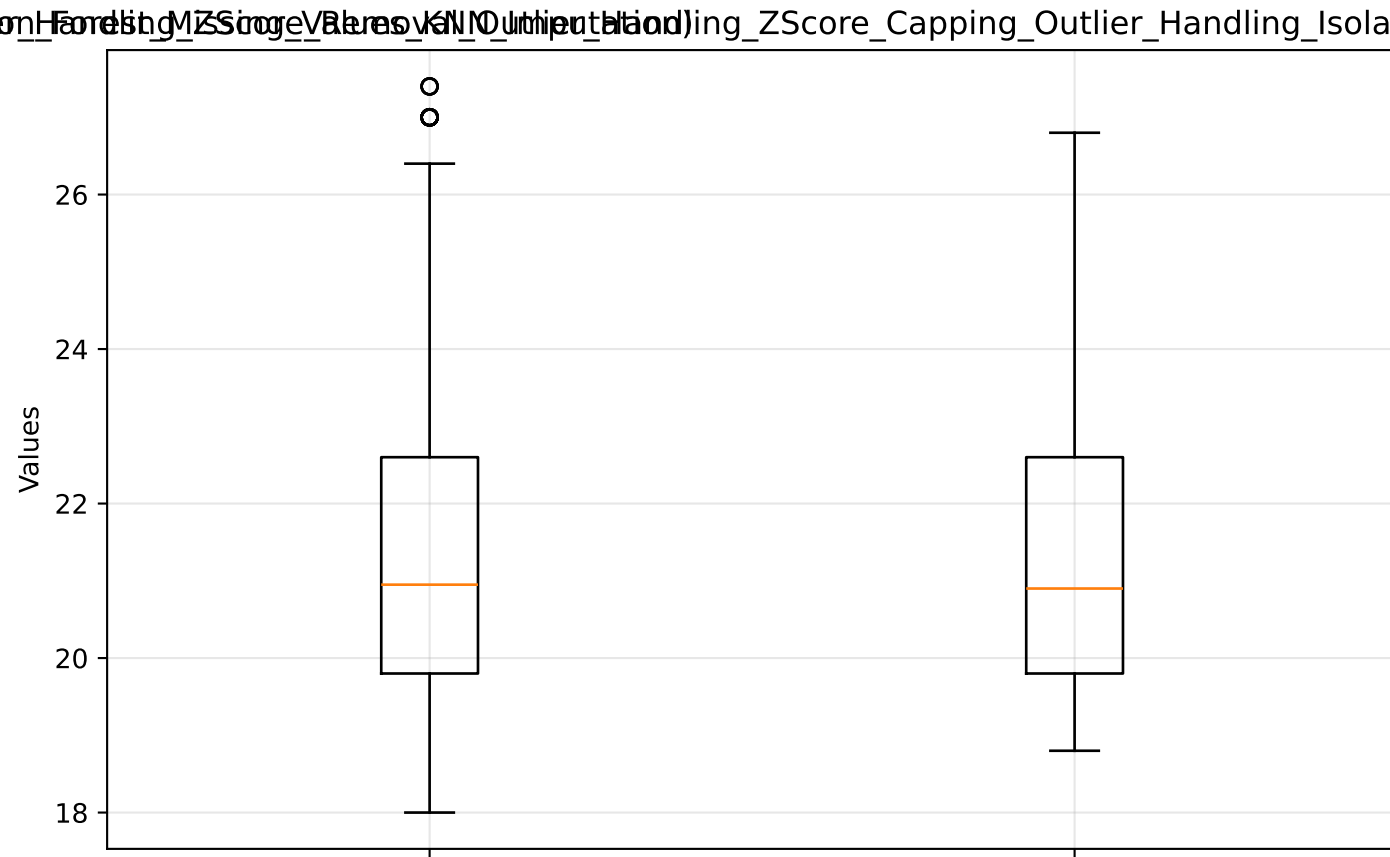
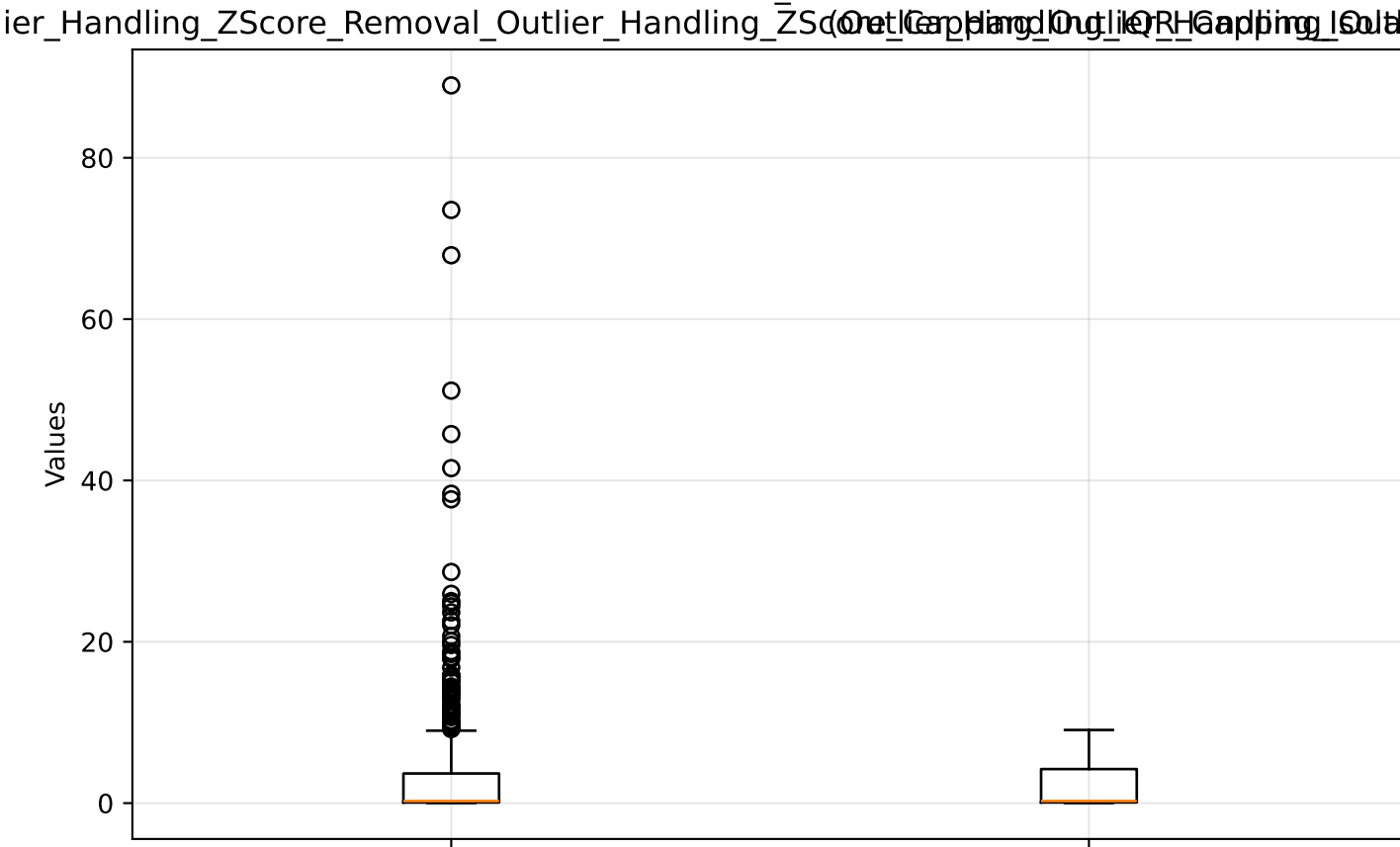
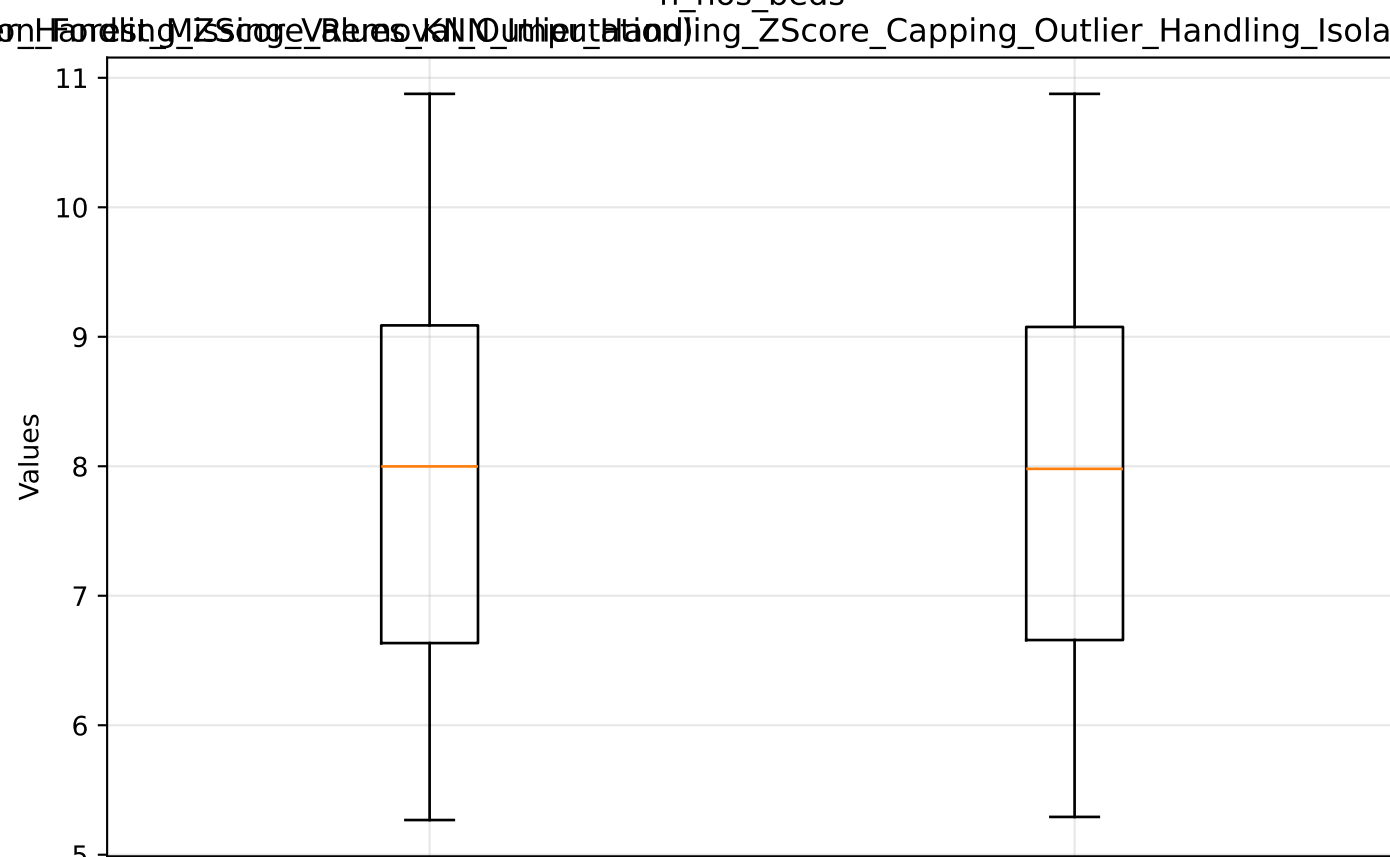
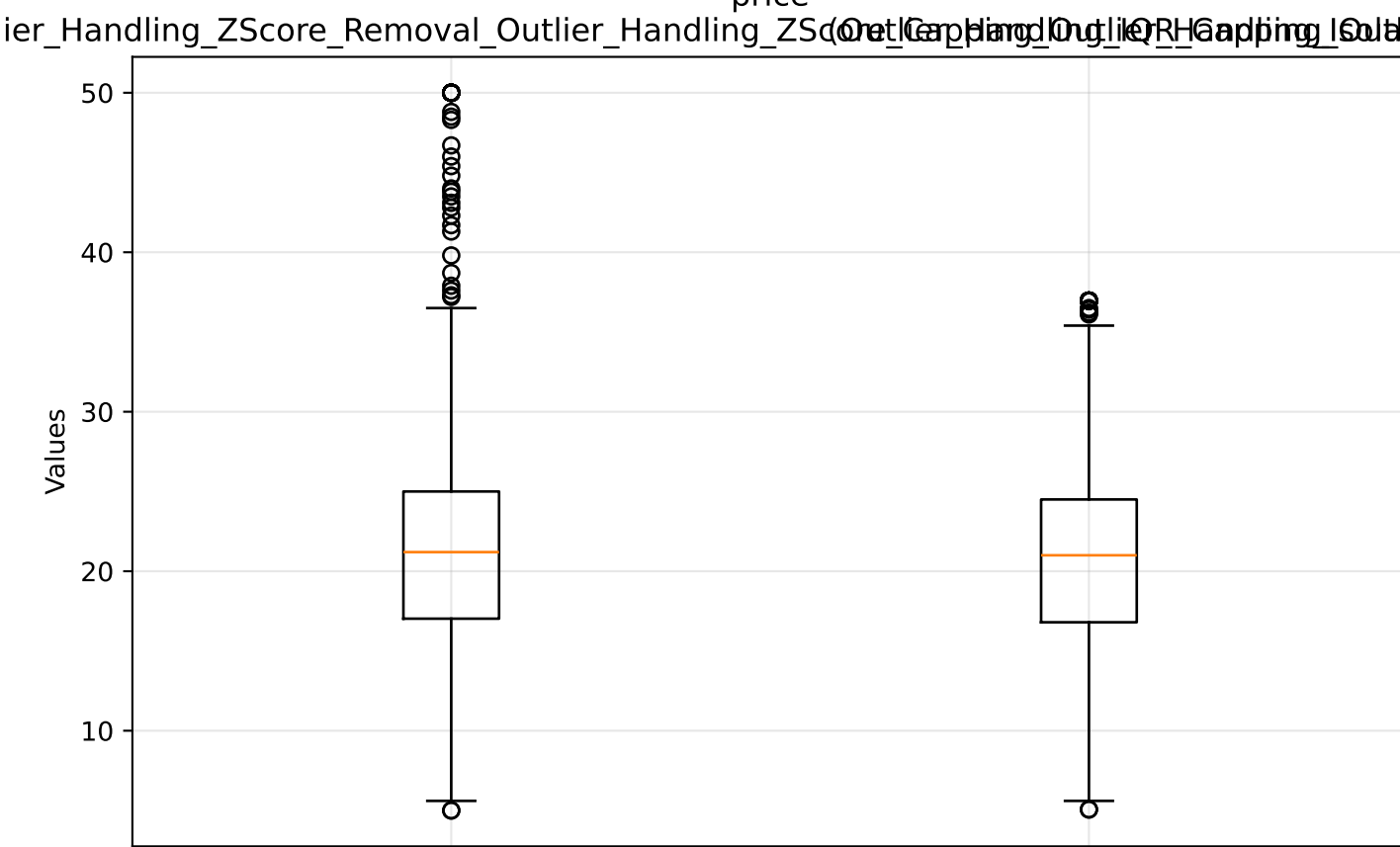
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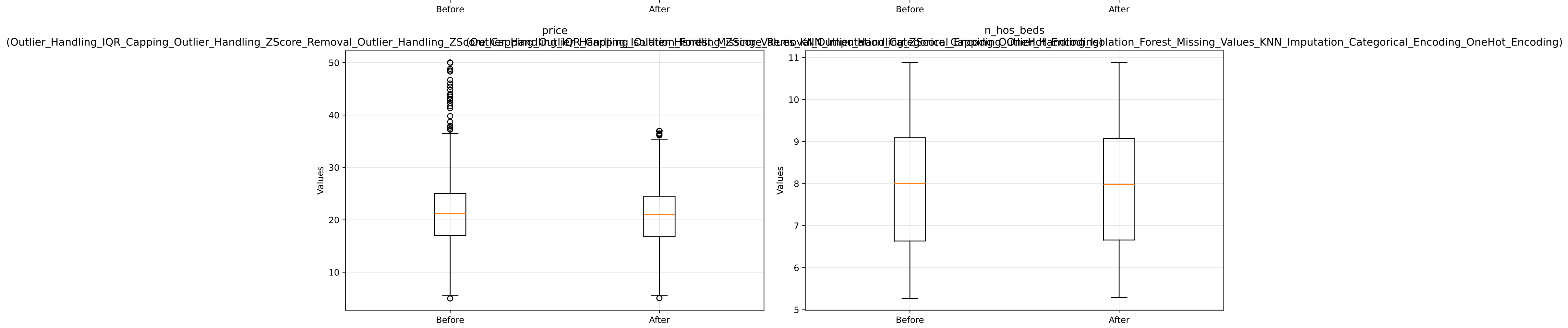
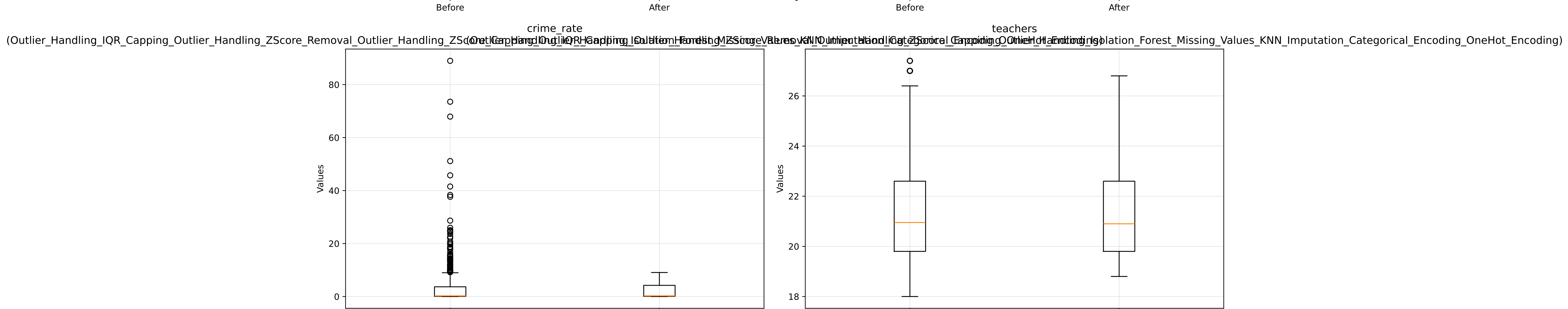
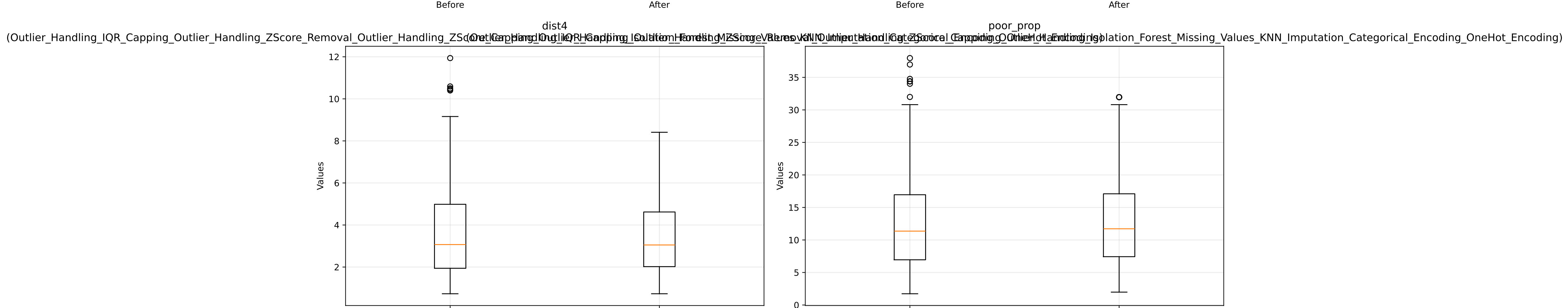
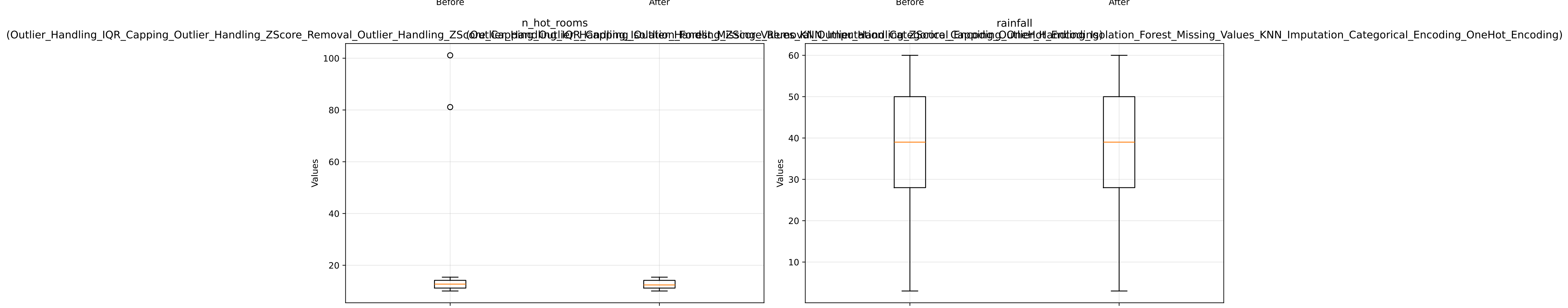
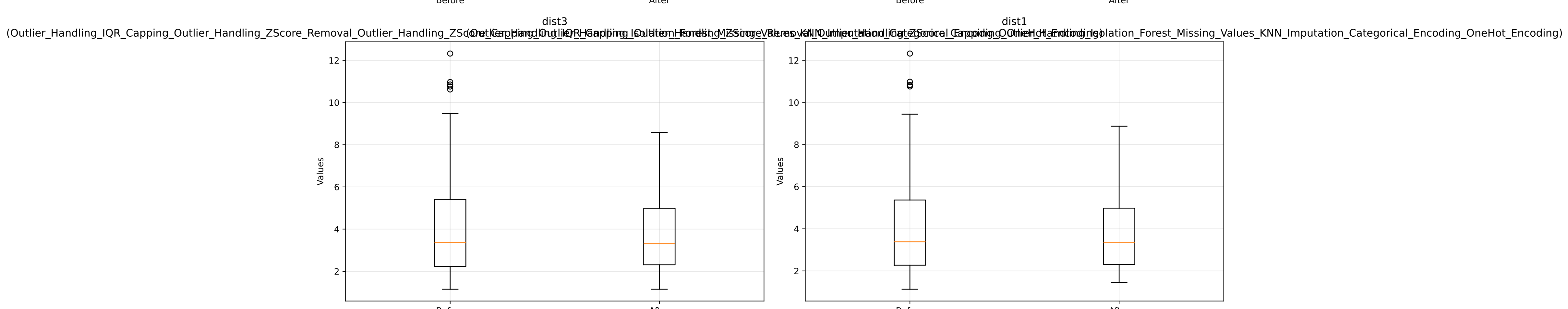
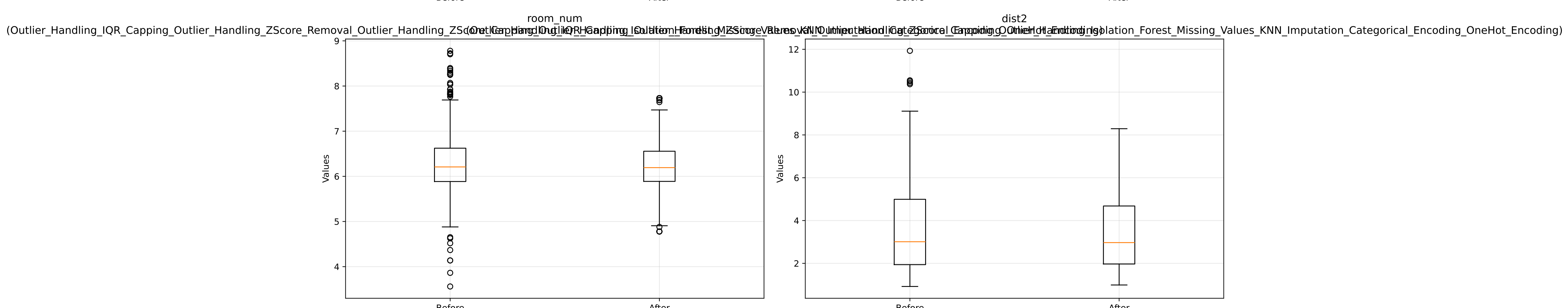
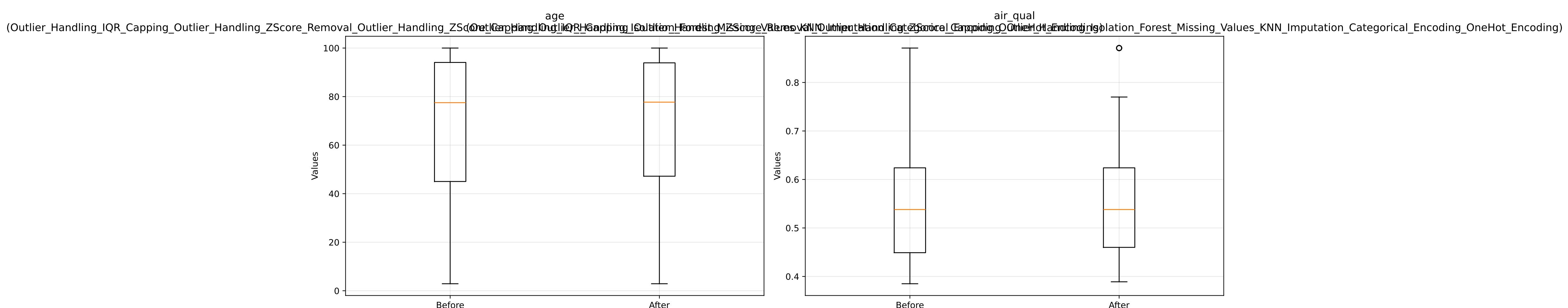
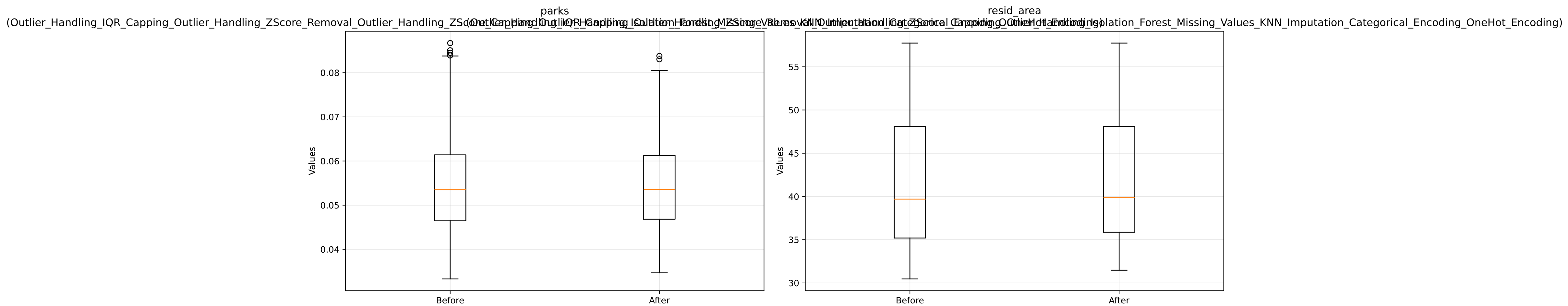
[illegible]

Before dist4 After Before poor_prop After



Before crime rate After Before teachers After

[illegible]



Combined Statistics - All Processing Steps

	Column	Method	Original Count	Cleaned Count	Original Mean	Cleaned Mean	Original Std	Cleaned Std	Removed Rows
	parks	Outlier Handling IQR Capping	506.0	506.0	0.0545	0.0544	0.0106	0.0106	0
	resid_area	Outlier Handling IQR Capping	506.0	506.0	41.1368	41.1368	6.8604	6.8604	0
	age	Outlier Handling IQR Capping	506.0	506.0	68.5749	68.5749	28.1489	28.1489	0
	air_qual	Outlier Handling IQR Capping	506.0	506.0	0.5547	0.5547	0.1159	0.1159	0
	room_hum	Outlier Handling IQR Capping	506.0	506.0	6.2848	6.2755	0.7026	0.6502	1
	dist2	Outlier Handling IQR Capping	506.0	506.0	3.6288	3.6182	2.1086	2.0728	1
	dist3	Outlier Handling IQR Capping	506.0	506.0	3.9607	3.9514	2.1198	2.0895	1
	dist1	Outlier Handling IQR Capping	506.0	506.0	3.972	3.9608	2.1085	2.0724	1
	n_hot_rooms	Outlier Handling IQR Capping	506.0	506.0	13.0416	12.7318	5.239	1.7173	2
	rainfall	Outlier Handling IQR Capping	506.0	506.0	39.1818	39.1818	12.0397	12.5302	2
	dist4	Outlier Handling IQR Capping	506.0	506.0	3.619	3.6076	2.0997	2.0537	2
	poor_prop	Outlier Handling IQR Capping	506.0	506.0	12.6531	12.612	7.1411	7.0168	2
	crime_rate	Outlier Handling IQR Capping	506.0	506.0	3.6135	3.6135	8.6015	3.3134	2
	teachers	Outlier Handling IQR Capping	506.0	506.0	21.5445	21.5362	7.1649	7.1439	2
	price	Outlier Handling IQR Capping	506.0	506.0	22.5289	21.8732	9.1822	7.5847	2
	n_hos_beds	Outlier Handling IQR Capping	506.0	506.0	7.8998	7.8998	1.4767	1.4767	2
	parks	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	0.0545	0.0544	0.0106	0.0106	2
	resid_area	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	41.1368	41.131	6.8604	6.8646	2
	age	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	68.5749	68.5569	28.1489	28.1932	2
	air_qual	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	0.5547	0.5547	0.1159	0.116	2
	room_hum	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	6.2848	6.2738	0.7026	0.659	2
	dist2	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	3.6288	3.6182	2.1086	2.0728	2
	dist3	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	3.9607	3.9525	2.1198	2.0919	2
	dist1	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	3.972	3.9625	2.1085	2.0742	2
	n_hot_rooms	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	13.0416	12.7318	5.239	1.68	2
	rainfall	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	39.1818	39.1818	12.0397	12.5302	2
	dist4	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	3.619	3.6076	2.0997	2.0537	2
	poor_prop	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	12.6531	12.609	7.1411	7.0042	2
	crime_rate	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	3.6135	3.6135	8.6015	3.3115	2
	teachers	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	21.5445	21.5383	7.1649	7.1466	2
	price	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	22.5289	21.8636	9.1822	7.5688	2
	n_hos_beds	Outlier Handling IQR Capping Outlier Handling	506.0	504.0	7.8998	7.8998	1.4767	1.4767	2
	parks	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.0545	0.0544	0.0106	0.0106	2
	resid_area	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	41.1368	41.131	6.8604	6.8646	2
	age	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	68.5749	68.5569	28.1489	28.1932	2
	air_qual	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.5547	0.5547	0.1159	0.116	2
	room_hum	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	6.2848	6.2738	0.7026	0.659	2
	dist2	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.6288	3.6182	2.1086	2.0633	2
	dist3	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.9607	3.9474	2.1198	2.0772	2
	dist1	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.972	3.9584	2.1085	2.0628	2
	n_hot_rooms	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	13.0416	12.7318	5.239	1.68	2
	rainfall	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	39.1818	39.1818	12.0397	12.5302	2
	dist4	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.619	3.6076	2.0997	2.0537	2
	poor_prop	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	12.6531	12.609	7.1411	7.0042	2
	crime_rate	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.6135	3.6135	8.6015	3.3115	2
	teachers	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	21.5445	21.5383	7.1649	7.1466	2
	price	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	22.5289	21.8636	9.1822	7.5688	2
	n_hos_beds	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	7.8998	7.8998	1.4767	1.4767	2
	parks	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.0545	0.0544	0.0106	0.0099	53
	resid_area	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	41.1368	41.3707	6.8604	6.6028	53
	age	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	68.5749	69.3349	28.1489	27.3738	53
	air_qual	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.5547	0.5547	0.1159	0.1077	53
	room_hum	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	6.2848	6.2474	0.7026	0.5871	53
	dist2	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.6288	3.4686	2.1086	1.7972	53
	dist3	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.9607	3.7998	2.1198	1.8117	53
	dist1	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.972	3.8118	2.1085	1.7998	53
	n_hot_rooms	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	13.0416	12.7301	5.239	1.6607	53
	rainfall	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	39.1818	39.1038	12.0397	12.4056	53
	dist4	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.619	3.4611	2.0997	1.7938	53
	poor_prop	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	12.6531	12.799	7.1411	6.7521	53
	crime_rate	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.6135	3.3258	8.6015	3.4046	53
	teachers	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	21.5445	21.3591	7.1649	7.1646	53
	price	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	22.5289	21.3421	9.1822	7.1364	53
	n_hos_beds	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	7.8998	7.8998	1.4767	1.454	53
	parks	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.0545	0.0544	0.0106	0.0099	53
	resid_area	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	41.1368	41.3707	6.8604	6.6028	53
	age	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	68.5749	69.3349	28.1489	27.3738	53
	air_qual	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.5547	0.5547	0.1159	0.1077	53
	room_hum	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	6.2848	6.2474	0.7026	0.5871	53
	dist2	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.6288	3.4686	2.1086	1.7972	53
	dist3	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.9607	3.7998	2.1198	1.8117	53
	dist1	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.972	3.8118	2.1085	1.7998	53
	n_hot_rooms	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	13.0416	12.7301	5.239	1.6607	53
	rainfall	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	39.1818	39.1038	12.0397	12.4056	53
	dist4	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.619	3.4611	2.0997	1.7938	53
	poor_prop	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	12.6531	12.799	7.1411	6.7521	53
	crime_rate	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.6135	3.3258	8.6015	3.4046	53
	teachers	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	21.5445	21.3591	7.1649	7.1646	53
	price	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	22.5289	21.3421	9.1822	7.1364	53
	n_hos_beds	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	7.8998	7.8998	1.4767	1.454	53
	parks	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.0545	0.0544	0.0106	0.0099	53
	resid_area	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	41.1368	41.3707	6.8604	6.6028	53
	age	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	68.5749	69.3349	28.1489	27.3738	53
	air_qual	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.5547	0.5547	0.1159	0.1077	53
	room_hum	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	6.2848	6.2474	0.7026	0.5871	53
	dist2	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.6288	3.4686	2.1086	1.7972	53
	dist3	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.9607	3.7998	2.1198	1.8117	53
	dist1	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.972	3.8118	2.1085	1.7998	53
	n_hot_rooms	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	13.0416	12.7301	5.239	1.6607	53
	rainfall	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	39.1818	39.1038	12.0397	12.4056	53
	dist4	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.619	3.4611	2.0997	1.7938	53
	poor_prop	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	12.6531	12.799	7.1411	6.7521	53
	crime_rate	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.6135	3.3258	8.6015	3.4046	53
	teachers	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	21.5445	21.3591	7.1649	7.1646	53
	price	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	22.5289	21.3421	9.1822	7.1364	53
	n_hos_beds	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	7.8998	7.8998	1.4767	1.454	53
	parks	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	0.0545	0.0544	0.0106	0.0099	53
	resid_area	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	41.1368	41.3707	6.8604	6.6028	53
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	dist3	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.9607	3.7998	2.1198	1.8117	53
	dist1	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	3.972	3.8118	2.1085	1.7998	53
	n_hot_rooms	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	13.0416	12.7301	5.239	1.6607	53
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	poor_prop	Outlier Handling IQR Capping Outlier Handling ZScore Removal	506.0	504.0	12.6531	12.799	7.1411	6.7521	53
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	dist1	Outlier Handling IQR Capping Outlier Handling ZScore Removal</							